

A Reduced WNT–HOX–Retinoic Acid Model of Colorectal Cancer Stemness and Its Approximation by Physics-Informed Neural Networks

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Abstract

Colorectal cancer initiation follows loss of APC function and the consequent deregulation of WNT/ β -catenin signaling, which retinoic acid (RA) signaling opposes through the HOX transcription factors. We formulate a reduced seven-state ordinary differential equation (ODE) model of the coupled WNT–RA–HOX network, nondimensionalize it to 36 kinetic parameters, and define four regimes — *Normal*, *Early Adenoma*, *Advanced Adenoma*, and *Severe APC Loss* — that reproduce the adenoma–carcinoma sequence by increasing WNT drive W while decreasing APC functionality θ_P . We then establish that this system admits an accurate physics-informed neural network (PINN) approximation in both directions. In the forward direction, a multi-scale Fourier-feature time embedding resolves the two-timescale retinoid forcing that defeats a standard multilayer perceptron through spectral bias, and reproduces stiff **Radau** reference trajectories to 1–3% relative error in every regime, including from only 40 sparse observations. In the inverse direction, parameter recovery saturates well below the full set, and we separate two causes: replacing the biased automatic-differentiation residual with a derivative-free integral residual roughly doubles the number of parameters recovered to within 10%, while the remaining ceiling is a property of the experiment and yields only to perturbations that excite otherwise-unobserved parts of the network. Global sensitivity analysis, Fisher information spectra, profile likelihood, and Hamiltonian Monte Carlo posteriors independently localize a genuine wall: in high-WNT regimes β -catenin saturates and θ_P becomes practically non-identifiable. For models of this class the binding constraint is the information content of the experiment, not the expressiveness of the network.

Keywords. Physics-informed neural networks; inverse problems; parameter identifiability; systems biology; colorectal cancer; WNT/ β -catenin signaling; retinoic acid; uncertainty quantification.

Highlights.

- A reduced seven-state WNT–RA–HOX ODE model of colorectal cancer stemness is derived and nondimensionalized to 36 parameters over four disease regimes.

- A Fourier-feature PINN solves the model forward to 1–3% error; spectral bias makes the embedding necessary, not optional.
- An integral (derivative-free) residual removes an estimator bias and roughly doubles inverse parameter recovery.
- Sensitivity analysis, Fisher information, profile likelihood, and Bayesian posteriors agree that β -catenin saturation renders APC functionality non-identifiable at high WNT drive.
- Experimental information, not network capacity, is the binding constraint on parameter recovery.

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1 Introduction

1.1 Biological background

Colorectal cancer follows one of the best-characterized progression sequences in oncology. The classical adenoma–carcinoma model of Fearon and Vogelstein [1] places biallelic inactivation of the *APC* tumor suppressor at the very beginning of the sequence, before *KRAS* activation and *TP53* loss. APC is a scaffolding component of the destruction complex that phosphorylates and targets β -catenin for proteasomal degradation; when APC function is lost, β -catenin accumulates, translocates to the nucleus, and forms a transcriptionally active complex with TCF/LEF [2, 3]. Among the direct targets of that complex is *MYC* [4], which couples WNT hyperactivation to proliferation and to the maintenance of a stem-like state. The centrality of this axis to intestinal homeostasis and to its neoplastic failure is well established [5, 6].

Retinoic acid, the biologically active metabolite of vitamin A, provides the principal differentiating counterweight. RA signaling drives the transcriptional programs that push progenitors out of the stem compartment [7], and its therapeutic use is not hypothetical: all-*trans* retinoic acid (ATRA) converted acute promyelocytic leukemia from a lethal diagnosis into a largely curable one [8]. The link from RA to the WNT axis in the colon runs through the HOX transcription factors. HOXA5 is a differentiation-promoting, RA-inducible factor that directly antagonizes WNT signaling and suppresses stem-cell traits in colorectal cancer [9]; the posterior HOX genes, HOXA13 among them, act in the opposite direction and are associated with a proliferative, stem-like phenotype [10]. RA levels are themselves autoregulated: RA induces the cytochrome P450 enzyme CYP26A1, which hydroxylates and inactivates RA, closing a negative feedback loop that limits the duration of any RA signal [11].

Combining these facts yields a compact regulatory picture built from three feedback structures:

$$\text{pro-stemness loop: } \beta\text{-catenin} \rightarrow \text{MYC} \rightarrow \text{HOXA13} \rightarrow \beta\text{-catenin}, \quad (1)$$

$$\text{differentiation loop: } \text{RA} \rightarrow \text{HOXA5} \rightarrow \text{APC} \dashv \beta\text{-catenin}, \quad (2)$$

$$\text{RA homeostasis loop: } \text{RA} \rightarrow \text{CYP26A1} \dashv \text{RA}. \quad (3)$$

The first is a positive feedback loop that amplifies WNT drive; the second is the negative feedback through which retinoid signaling restrains it; the third is the autoregulation that makes retinoid signaling transient. The model developed in Section 2 is the minimal ODE system carrying all three, with WNT drive W and APC functionality θ_P as the two knobs that move the system along the adenoma–carcinoma sequence.

1.2 Rationale for a physics-informed neural network approach

Mechanistic models of signaling pathways of this kind have a long history [12], and the forward problem — integrating the ODEs given the parameters — is essentially solved by mature stiff solvers. The difficulty lies in the inverse direction: given sparse, noisy trajectory measurements,

recover the kinetic parameters. This is a classically ill-posed problem in systems biology [13–15], and for models of this size it is routinely under-determined regardless of the estimation method used.

Physics-informed neural networks [16, 17] offer an attractive formulation for exactly this setting. A network $z_\phi(\tau)$ is trained to represent the state trajectory, and the governing equations enter the loss as a residual penalty evaluated at collocation points rather than as a hard constraint discharged by a time-stepper. Unknown parameters θ are then simply additional trainable variables optimized jointly with the network weights [16]. Because the residual is enforced everywhere in the domain and not only where data exist, the approach suits sparse and irregular sampling, it is supported by mature software [18], and it extends naturally to systems-biology models [19, 20]. The framework is also a natural home for uncertainty quantification: replacing point estimates by a posterior over parameters and weights gives Bayesian PINNs [21], from which per-parameter credible intervals — and therefore an honest identifiability verdict — follow directly.

The framework is not free of pathologies. PINNs are known to fail on problems with sharp features or multiple timescales [22], and the failure has been traced both to gradient imbalance among loss terms [23] and to the spectral bias of the underlying network, which learns low frequencies far faster than high ones [24, 25]. Stiff chemical kinetics is the canonical hard case [26]. Our model contains exactly the ingredient that triggers this: a circadian RA input of period $T_R = 24$ superimposed on a treatment pulse spanning tens of time units, over a horizon of 150. Fourier-feature embeddings are the standard remedy [27], and are what we adopt.

1.3 Scope and contributions

This paper has two objects: a new mechanistic model of the WNT–RA–HOX axis, and a demonstration that this model can be approximated by physics-informed neural networks in both the forward and the inverse direction, together with a characterization of where that approximation stops working and why. It deliberately stops short of the neural-mechanistic hybrid (universal differential equation) extensions, which are the subject of separate work. Its contributions are:

1. **A reduced seven-state WNT–RA–HOX model** of colorectal cancer stemness (Section 2), nondimensionalized to 36 parameters, with four regimes spanning the adenoma–carcinoma sequence, a circadian-plus-pulse retinoid forcing, and a derived stemness index as the scientific readout.
2. **A three-method global sensitivity analysis** (Section 4) — local elasticity, Morris elementary effects, and Sobol variance decomposition — applied both to the reduced model and to the full 27-state reaction network from which it descends, producing a cross-method consensus ranking and an explicit list of parameters that can be fixed without loss.
3. **A forward PINN** (Section 5) that solves the system across all four regimes. We document the spectral-bias failure of a plain multilayer perceptron on the two-timescale forc-

ing and the multi-scale Fourier-feature embedding that removes it, and validate against a `solve_ivp`/Radau reference at $\text{rtol} = 10^{-10}$, $\text{atol} = 10^{-12}$.

4. **A systematic study of inverse recovery** (Section 6), scaling from a two-parameter proof of concept to the full 36-parameter problem across four regimes and multiple experimental conditions. We isolate two distinct causes of the observed recovery ceiling and address them separately: an *estimator* defect, cured by replacing the automatic-differentiation residual with a derivative-free integral residual; and an *information* deficit, addressed by adding parameter-exciting perturbation conditions. We also report a negative result — shrinking and regularizing the network does not help — which we regard as informative.
5. **A three-lens identifiability verdict** (Section 7): Fisher information spectra, profile likelihood, and HMC posteriors independently agree that θ_P becomes non-identifiable as WNT drive increases, and that the recoverable information concentrates in a small well-conditioned subset of parameters.
6. **A reproducible calibration failure** in the Bayesian analysis (Section 8), in which a physics log-likelihood summed over 2.1×10^5 residual terms produced a pinpoint posterior that excluded the true parameter values by a median of 29–51 standard deviations while reporting every parameter as identifiable. We report it because the failure mode is generic to Bayesian PINNs and, in our reading, not adequately flagged in the literature.

2 Model formulation

2.1 Biological assumptions and network topology

The model encodes the following assumptions, each drawn from the literature surveyed above. WNT signaling promotes β -catenin accumulation. β -catenin activates MYC, HOXA13, and CYP26A1. Retinoic acid activates HOXA5. HOXA5 promotes APC expression and suppresses β -catenin. APC negatively regulates β -catenin. CYP26A1 degrades retinoic acid. In summary, WNT signaling is the stimulus that promotes accumulation of β -catenin, MYC, HOXA13, and CYP26A1, the species associated with proliferation; retinoic acid is the stimulus that promotes accumulation of HOXA5 and APC, the species associated with differentiation and tumor suppression.

The human body does not synthesize vitamin A *de novo*, so retinoid input must be obtained through diet. We therefore model dietary RA input with a cosine term representing intermittent intake across the day, with circadian period $T_R = 24$. ATRA treatment, by contrast, is administered continuously over a defined therapeutic window; we model it with a difference of hyperbolic tangents, which gives a smooth pulse that switches on near τ_1 and off near τ_2 with controllable steepness q . The smoothness matters numerically as well as biologically: a discontinuous dose would place a non-differentiable point inside the domain over which the PINN residual is differentiated.

2.2 Governing equations

Let $Y = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$ denote the state vector, where B is β -catenin, P is APC, H_5 is HOXA5, H_{13} is HOXA13, M is MYC, R is retinoic acid, and C is CYP26A1. The dimensional model is

$$\frac{dB}{dt} = a_B W + v_{13} \frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} P B - v_5 \frac{H_5 B}{K_5 + B}, \quad (4)$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - (d_{P0} + d_{P1}(1 - \Theta_P)) P, \quad (5)$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M}, \quad (6)$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13} \frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13} \frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13}, \quad (7)$$

$$\frac{dM}{dt} = a_M + v_{BM} \frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M, \quad (8)$$

$$\begin{aligned} \frac{dR}{dt} = & u_0 + A_d \left[1 + \cos \left(\frac{2\pi t}{T_d} - \phi \right) \right] + \frac{D}{2} \left[\tanh(q(t - t_1)) - \tanh(q(t - t_2)) \right] \\ & - d_R R - k_{CR} C R, \end{aligned} \quad (9)$$

$$\frac{dC}{dt} = a_C + v_{RC} \frac{R}{K_{RC} + R} + v_{BC} \frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C. \quad (10)$$

Structure of the equations. Despite their length, Eqs. (4)–(10) are seven instances of a single elementary statement. Each is a mass balance for one species,

$$\underbrace{\frac{d(\text{species})}{dt}}_{\text{rate of change}} = \underbrace{\text{production}}_{\text{input}} - \underbrace{\text{loss}}_{\text{output}}, \quad (11)$$

that is, $\text{rate} = \text{input} - \text{output}$: a species accumulates exactly as fast as it is made minus as fast as it is removed. Every term entering with a plus sign is a route of synthesis (basal expression, activation by an upstream regulator, or, for RA, external supply), and every term entering with a minus sign is a route of removal (spontaneous turnover, enzymatic degradation, or sequestration by a partner). All of the biology sits in two choices: which species appear in which term, and what functional form each term takes. We use Hill functions $x^n/(\kappa^n + x^n)$ where the regulation is cooperative and transcriptional ($n_B = n_M = n_H = 2$), Michaelis–Menten saturation $x/(\kappa + x)$ where it represents an enzymatic or titration process, and mass-action products xy where two species must physically meet. Each of these saturates, so no production term can grow without bound — the biologically necessary property that concentrations remain finite. Table 1 reads the seven equations back in this form.

State	Eq.	Input (production)	Output (loss)
B (β -catenin)	(4)	WNT drive; HOXA13 feedback	turnover; APC-mediated degradation; HOXA5 suppression
P (APC)	(5)	expression, promoted by HOXA5 and inhibited by β -catenin/HOXA13	turnover, accelerated by loss of APC functionality Θ_P
H_5 (HOXA5)	(6)	basal expression; RA induction	turnover; MYC-mediated suppression
H_{13} (HOXA13)	(7)	basal expression; β -catenin and MYC induction	turnover
M (MYC)	(8)	basal expression; β -catenin induction	turnover
R (RA)	(9)	basal supply; circadian dietary intake; ATRA treatment pulse	turnover; CYP26A1-mediated catabolism
C (CYP26A1)	(10)	basal expression; RA and β -catenin induction	turnover

Table 1: Each governing equation as an input–output balance, Eq. (11). The columns list the source and sink terms of the corresponding equation in words; the pathway structure of Section 2 is recovered by reading the table row by row.

Every biological assumption stated above appears explicitly in this bookkeeping. In Eq. (4), WNT signaling W drives β -catenin synthesis while APC contributes to its degradation through the mass-action term $k_{PB}PB$, encoding the inverse relationship between the two: the two species must meet for degradation to occur, so the loss rate is proportional to the product of their concentrations. In Eq. (5), Θ_P is APC functionality: at $\Theta_P = 1$ the excess degradation term vanishes and APC turns over at its baseline rate d_{P0} , while decreasing Θ_P raises the effective degradation rate and so models progressive loss of APC function.

2.3 Nondimensionalization and scaling

To reduce the number of independent parameters and improve the conditioning of both the forward solve and the training problem, we nondimensionalize. Each concentration is scaled by its characteristic value, taken to be its initial condition, and time is scaled by the reference β -catenin degradation rate $d_B = 1 \text{ hr}^{-1}$:

$$B = B_0 b, \quad P = P_0 p, \quad H_5 = H_{5,0} h_5, \quad H_{13} = H_{13,0} h_{13}, \quad M = M_0 m, \quad R = R_0 r, \quad C = C_0 c, \quad (12)$$

with $\tau = d_B t$. Scaling time by d_B is the natural choice because β -catenin is directly activated by WNT signaling and directly affects most other species, so its turnover sets the reference timescale

of the network. The resulting dimensionless system is

$$\frac{db}{d\tau} = W + \eta_{13} \frac{h_{13}^{n_H}}{\kappa_{13}^{n_H} + h_{13}^{n_H}} - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}, \quad (13)$$

$$\frac{dp}{d\tau} = \frac{1}{\varepsilon_P} \left[\frac{1 + \rho_5 h_5}{1 + \rho_B b + \rho_{13} h_{13}} - \delta_P(\theta_P) p \right], \quad (14)$$

$$\frac{dh_5}{d\tau} = \frac{1}{\varepsilon_5} \left[a_5 + \eta_R \frac{r}{\kappa_R + r} - h_5 - \eta_M \frac{m h_5}{\kappa_M + m} \right], \quad (15)$$

$$\frac{dh_{13}}{d\tau} = \frac{1}{\varepsilon_{13}} \left[a_{13} + \eta_{B13} \frac{b^{n_B}}{\kappa_{B13}^{n_B} + b^{n_B}} + \eta_{M13} \frac{m^{n_M}}{\kappa_{M13}^{n_M} + m^{n_M}} - h_{13} \right], \quad (16)$$

$$\frac{dm}{d\tau} = \frac{1}{\varepsilon_M} \left[a_M + \eta_{BM} \frac{b^{n_B}}{\kappa_{BM}^{n_B} + b^{n_B}} - m \right], \quad (17)$$

$$\frac{dr}{d\tau} = \frac{1}{\varepsilon_R} \left[\mu_R(\tau) - r - \lambda_C c r \right], \quad (18)$$

$$\frac{dc}{d\tau} = \frac{1}{\varepsilon_C} \left[a_C + \eta_{RC} \frac{r}{\kappa_{RC} + r} + \eta_{BC} \frac{b^{n_B}}{\kappa_{BC}^{n_B} + b^{n_B}} - c \right]. \quad (19)$$

The balance structure of Eq. (11) is preserved term by term: each bracket is still input minus output, with every linear loss term normalized to $-x$ by the choice of scaling, and the prefactor $1/\varepsilon_X$ is a pure timescale ratio saying how fast species X relaxes relative to β -catenin. Nondimensionalization has therefore not changed the model, only its units and its parameter count, which falls to the 36 dimensionless groups of Table 2.

The APC degradation modifier is

$$\delta_P(\theta_P) = 1 + \delta_{P1}(1 - \theta_P), \quad (20)$$

so $\theta_P = 1$ recovers normal turnover and $0 < \theta_P < 1$ represents an APC-mutant cell. The retinoid forcing is

$$\mu_R(\tau) = \mu_0 + A_R \left[1 + \cos\left(\frac{2\pi\tau}{T_R} - \phi\right) \right] + \frac{D_R}{2} \left[\tanh(q(\tau - \tau_1)) - \tanh(q(\tau - \tau_2)) \right], \quad (21)$$

combining a constant basal supply μ_0 , the circadian dietary term of amplitude A_R and period T_R , and the smooth ATRA treatment pulse of amplitude D_R over the window $[\tau_1, \tau_2]$. The ε coefficients are relative timescale ratios; several are well below unity ($\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$), which is the source of the mild stiffness discussed in Section 3.

The scientific readout is the dimensionless *stemness index*

$$S(\tau) = \frac{b(1 + \alpha_{13} h_{13})}{(1 + p)(1 + \alpha_5 h_5)}, \quad \alpha_5 = \alpha_{13} = 1, \quad (22)$$

which is large when the proliferative species dominate and small when the differentiating species

do. It is not a state variable; it is a derived quantity tracked alongside the states because it is the aggregate the model exists to predict.

2.4 Parameter values and disease regimes

Table 2 lists the dimensionless kinetic parameters and the forcing parameters at the baseline values used throughout. Initial conditions are $y_0 = (0.20, 1.00, 0.80, 0.30, 0.30, 0.60, 0.40)$, and the integration horizon is $T = 150$, chosen to cover the active-dynamics window including the full treatment pulse and the relaxation that follows it.

Equation	Parameter	Value	Role
b (13)	W	regime	WNT drive
	η_{13}	0.75	HOXA13 $\rightarrow$ β -catenin activation
	κ_{13}	0.55	half-saturation
	λ_P	1.60	APC-mediated degradation
	λ_5	1.30	HOXA5-mediated suppression
	κ_5	0.50	half-saturation
p (14)	θ_P	regime	APC functionality
	ε_P	1.00	timescale
	ρ_5	1.10	HOXA5 $\rightarrow$ APC induction
	ρ_B	1.10	β -catenin repression of APC
	ρ_{13}	1.30	HOXA13 repression of APC
	δ_{P1}	3.50	excess degradation on APC loss
h_5 (15)	ε_5	1.20	timescale
	a_5	0.15	basal production
	η_R	2.50	RA $\rightarrow$ HOXA5 activation
	κ_R	0.40	half-saturation
	η_M	2.50	MYC repression of HOXA5
	κ_M	0.50	half-saturation
h_{13} (16)	ε_{13}	1.00	timescale
	a_{13}	0.18	basal production
	η_{B13}	0.95	β -catenin $\rightarrow$ HOXA13
	κ_{B13}	0.50	half-saturation
	η_{M13}	0.55	MYC $\rightarrow$ HOXA13
	κ_{M13}	0.50	half-saturation
m (17)	ε_M	0.60	timescale
	a_M	0.18	basal production
	η_{BM}	1.35	β -catenin $\rightarrow$ MYC
	κ_{BM}	0.50	half-saturation
r (18)	ε_R	0.40	timescale
	λ_C	0.85	CYP26A1-mediated RA degradation
c (19)	ε_C	0.80	timescale
	a_C	0.08	basal production
	η_{RC}	1.50	RA $\rightarrow$ CYP26A1
	κ_{RC}	0.50	half-saturation
	η_{BC}	1.50	β -catenin $\rightarrow$ CYP26A1
	κ_{BC}	0.50	half-saturation
forcing (21)	μ_0	0.35	basal RA supply
	A_R	0.04	circadian amplitude
	T_R	24.0	circadian period
	ϕ	0.0	phase
	D_R	1.50	ATRA dose
	q	0.30	pulse steepness
	τ_1, τ_2	40.0, 88.0	treatment window
Hill	n_B, n_M, n_H	2	cooperativity (fixed)

Table 2: Dimensionless parameters at baseline. The 36 kinetic parameters (everything above the forcing block, with W and θ_P set per regime) form the unknown vector in the inverse problem of Section 6. Hill exponents are held fixed at 2 throughout.

Disease progression is modeled by simultaneously increasing WNT drive and decreasing APC functionality, mimicking the adenoma–carcinoma sequence. This gives the four regimes of Table 3.

Regime	W	θ_P	Interpretation
Normal	0.80	1.00	healthy tissue, low WNT
Early Adenoma	1.00	0.75	early lesion
Advanced Adenoma	1.50	0.50	elevated WNT, substantial APC loss
Severe APC Loss	2.00	0.25	saturated WNT, loss of APC control

Table 3: The four regimes. All other parameters are held at the baseline values of Table 2; the regimes differ only in (W, θ_P) .

It is worth stating plainly at the outset that this construction is also the source of the central identifiability difficulty of Section 7. The regimes sweep θ_P downward exactly as W is swept upward, and increasing W drives β -catenin toward saturation of the Hill terms that read it. In the saturated limit the trajectory becomes insensitive to how much APC functionality remains, so θ_P and W become entangled in precisely the regimes of greatest clinical interest.

3 Numerical reference solution

Ground-truth trajectories are computed with SciPy’s `solve_ivp` [28] using the implicit Radau method at $\text{rtol} = 10^{-10}$ and $\text{atol} = 10^{-12}$. The implicit solver is required rather than merely convenient: the spread in ε coefficients (Table 2) combined with the fast circadian forcing makes the system mildly stiff, and explicit methods either take prohibitively small steps or fail to resolve the oscillation. The tolerances are set several orders of magnitude tighter than any error we intend to report, so the reference can be treated as exact when scoring PINN accuracy.

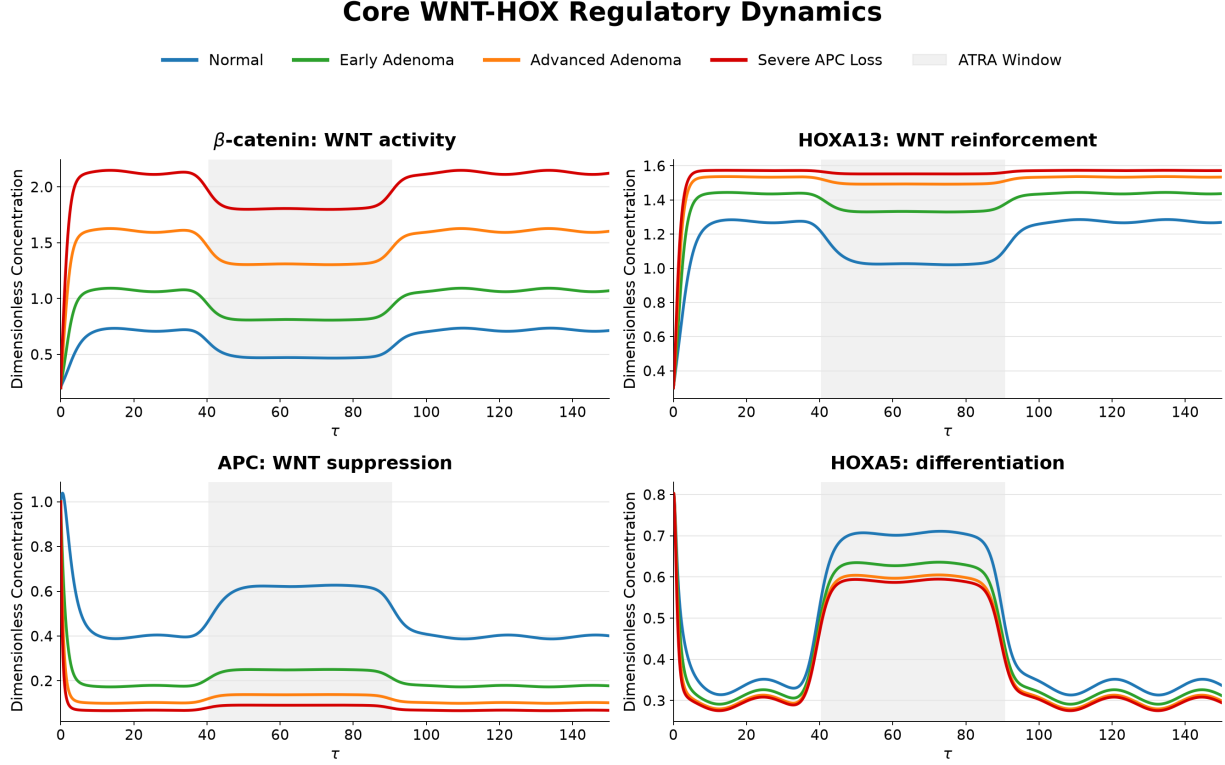


Figure 1: Reference trajectories of all seven states across the four regimes, computed with `solve_ivp/Radau`. The ATRA treatment window is $[\tau_1, \tau_2] = [40, 88]$.

The reference solutions in Figure 1 reproduce the qualitative behavior the model was constructed to capture. The concentrations of β -catenin and HOXA13 both fall during the ATRA treatment window, consistent with their role as proliferation-associated species, and both sit at higher levels in the more severe regimes. The HOXA13 trajectories are particularly informative: in the Advanced Adenoma and Severe APC Loss regimes the concentration barely decreases during treatment. The model therefore predicts that at later stages of progression ATRA becomes ineffective at suppressing HOXA13 — a loss of treatment response that emerges from the network rather than being imposed. Conversely, APC and HOXA5 rise during the treatment window, confirming their differentiating role, and both decline as the cancer worsens. While HOXA5 responds strongly to ATRA in all four regimes, APC is markedly less responsive at later stages, because the elevated degradation $\delta_P(\theta_P)$ caps the level APC can reach regardless of how much HOXA5 is induced.

4 Global sensitivity analysis

Before asking which parameters can be *recovered*, it is worth asking which parameters *matter*. A parameter to which no observable is sensitive cannot be recovered by any method, and identifying such parameters in advance both explains later failures and suggests a reduced model. We therefore screen the parameter space with three complementary methods, following standard practice in

systems biology [29], and we apply the same suite twice: to the reduced seven-state model that is the subject of this paper, and to the full 27-state reaction network from which it descends.

4.1 Sampling design and sensitivity measures

All three methods score the same class of quantity of interest: the time-averaged value of an output over the horizon,

$$Q[s] = \frac{1}{T} \int_0^T s(\tau) d\tau, \quad (23)$$

evaluated for each of the seven states and for the stemness index (22). Averaging rather than sampling at a point avoids making the ranking an artifact of an arbitrarily chosen readout time.

Local elasticity. The normalized local sensitivity

$$E_{ij} = \frac{\partial Q_i}{\partial p_j} \cdot \frac{p_j}{Q_i} \quad (24)$$

is computed by central differences with a relative step of 5%. It is signed, which makes it interpretable — the sign tells you the direction of the effect — but it is a linearization valid only in a neighborhood of the baseline.

Morris elementary effects. The Morris screening design [30, 31] samples one-at-a-time trajectories through a discretized parameter box and reports, for each parameter, the mean absolute elementary effect μ^* (overall influence) and the standard deviation σ (evidence of nonlinearity or interaction). The conventional flag $\sigma > \mu^*/2$ marks a parameter whose effect is strongly nonlinear or depends on the values of others. Morris is cheap — $N(P+1)$ model evaluations — and is used as a screen rather than a quantitative decomposition.

Sobol variance decomposition. The Sobol indices [29, 32] decompose the variance of Q induced by the parameter distribution into contributions from each parameter alone (S_1 , first order) and from each parameter including all its interactions (S_T , total order). A large gap $S_T \gg S_1$ is the signature of interaction; $\sum_j S_1^{(j)} \approx 1$ indicates a nearly additive response. Sobol is the most informative and the most expensive of the three.

All sampling and index estimation use SALib [33], and every parameter is varied within a $\pm 30\%$ box around its baseline value.

4.2 Sensitivity of the reduced seven-state model

For the reduced model we vary the 14 parameters that carry the regulatory couplings and the treatment: θ_P , D_R , A_R , η_R , η_M , η_{13} , η_{B13} , η_{BM} , η_{BC} , λ_C , λ_P , λ_5 , ρ_B , ρ_{13} . Morris uses $N = 40$ trajectories at 4 levels (600 model solves); Sobol uses $N = 256$ base samples without second-

order indices (4096 solves). Reference trajectories are integrated with LSODA at $\text{rtol} = 10^{-7}$, $\text{atol} = 10^{-9}$.

A sampling artifact, and its correction. One result must be reported before any other, because it invalidates part of the naive analysis. The $\pm 30\%$ box places θ_P in $[0.70, 1.30]$. But the APC degradation modifier (20) satisfies

$$\delta_P(\theta_P) = 1 + \delta_{P1}(1 - \theta_P) < 0 \quad \Longleftrightarrow \quad \theta_P > 1 + \frac{1}{\delta_{P1}} = 1.2857, \quad (25)$$

for $\delta_{P1} = 3.5$. Samples with $\theta_P > 1.2857$ therefore give APC a *negative* decay rate and the trajectory diverges. At $\theta_P = 1.30$ the time-averaged APC readout is 6.4×10^3 against a baseline of 0.473.

The contamination is not confined to APC. Because APC appears in the denominator of the stemness index (22), a diverging p collapses stemness to 0.002 at $\theta_P = 1.29$, so the stemness readout is corrupted by the same samples. In the raw output the damage is plainly visible: the Morris μ^* for APC reaches 5.2×10^3 with $\sigma = 5.1 \times 10^3$, and the Sobol first-order indices for APC sum to $1.145 > 1$ with $S_1(\theta_P) = 1.146 \pm 1.198$ — an impossible value for a variance share.

Since θ_P is a functionality fraction, $\theta_P > 1$ has no biological meaning in any case, so we re-ran the entire analysis with the θ_P box clipped at 1.0. This removes the divergence: the APC first-order indices then sum to 0.807, and no readout is anomalous. It also, unavoidably, halves the sampled range of θ_P (from 0.6 wide to 0.3) while leaving every other parameter at the full $\pm 30\%$. The clipped design is therefore no longer balanced, and θ_P 's global scores are not on an equal footing with the others. We report both boxes, compare *rankings* rather than magnitudes across them, and treat the clipped run as the valid one. Local elasticities, which perturb by only 5% and never leave the admissible region, are identical under both and are unaffected throughout.

We dwell on this because the hazard is general. A proportional-perturbation box is applied mechanically to every parameter at once, and it takes a deliberate check to notice that one of them has a hard constraint inside the box. The symptom here — a variance share exceeding one — was visible in the output, but only because we looked; the Morris ranking that the contaminated run produced was superficially plausible, and, as the next paragraph shows, wrong.

Results. Table 4 gives the three methods' verdicts on the stemness index, the aggregate readout the model exists to predict.

Parameter	elasticity	Morris μ^*	Sobol S_1	Sobol S_T
θ_P	-3.609	0.448	0.154	0.183
η_{13}	+1.038	0.533	0.232	0.247
η_R	-0.891	0.442	0.167	0.174
λ_P	-0.781	0.288	0.070	0.072
η_{B13}	+0.676	0.371	0.109	0.115
η_M	+0.645	0.333	0.086	0.093
λ_5	-0.512	0.258	0.064	0.064
ρ_{13}	+0.479	0.178	0.021	0.027
η_{BM}	+0.343	0.161	0.016	0.022
λ_C	+0.259	0.123	0.017	0.016
ρ_B	+0.219	0.082	0.006	0.008
η_{BC}	+0.160	0.082	0.006	0.007
D_R	-0.083	0.041	0.000	0.002
A_R	-0.037	0.018	0.002	0.000

Table 4: Sensitivity of the stemness index in the reduced model, ordered by elasticity magnitude. Morris and Sobol columns are from the θ_P -clipped sampling box; elasticities are box-independent. Recall that θ_P ’s clipped range is half the width of every other parameter’s, which depresses its two global scores relative to the rest of the column.

Four observations follow, one of which is a correction.

First, **APC functionality dominates the local picture**. Its elasticity on stemness is -3.61 , three and a half times the next largest, and the sign is correct — more APC function, less stemness. The same parameter has the largest elasticity on every individual state, positive on APC ($+4.85$) and negative on β -catenin (-2.01).

Second, and this is the correction, **that dominance does not survive into the global measures once the sampling artifact is removed**. In the contaminated box θ_P towered over everything, with Morris $\mu^* = 1.212$ against 0.337 for the runner-up and $S_1 = 0.750$ of stemness variance. In the clipped box its Morris score falls to 0.448 , slightly *below* η_{13} at 0.533 , and its S_1 falls to 0.154 , below both η_{13} (0.232) and η_R (0.167). The overall Morris ranking across all eight outputs likewise changes from θ_P alone at the top to a cluster of η_R (0.509), η_{13} (0.497) and θ_P (0.478). Part of that drop is the halved sampling range and is an artifact of the fix rather than a finding; but the direction is unambiguous, and the honest statement is that θ_P sits in a top cluster of three or four parameters globally rather than standing alone. A conclusion drawn from the uncorrected run — and it is the conclusion one would naturally have drawn — would have overstated the case.

The two pictures are not in conflict, because they answer different questions. Local elasticity asks how strongly the output responds to a small perturbation *at the baseline operating point*, where $\theta_P = 1$ and the APC degradation term is exactly at the hinge of Eq. (20). Morris and Sobol ask how much of the output variation is attributable to a parameter *across a whole range*. A parameter can be the steepest lever locally and still account for a modest share of variance over a box, which

is precisely θ_P 's situation.

Third, **each state has an identifiable own-pathway driver**. The second-largest elasticity on HOXA5 is η_R (+1.02, the RA→HOXA5 arm); on HOXA13 it is η_{B13} (+0.61); on MYC it is η_{BM} (+0.93); on CYP26A1 it is η_{BC} (+0.57); and on RA it is λ_C (−0.55) and D_R (+0.47), the CYP26A1 degradation arm and the treatment dose. The network's wiring is legible in the sensitivity structure.

Fourth, and most useful for what follows, **the reduced model is essentially additive**. For stemness in the clipped box $\sum_j S_1 = 0.950$ and $\sum_j S_T = 1.029$: 95% of the output variance is first-order, and the largest gap $S_T - S_1$ over the whole table is 0.016. Removing the divergent samples also cleans up the Morris nonlinearity diagnostic — in the contaminated box 13 of 14 parameters were flagged $\sigma > \mu^*/2$, whereas in the clipped box *none* is. What looked like pervasive nonlinearity was the divergence. This matters for the rest of the paper: when parameter recovery fails in Section 6, the failure is not caused by a tangle of high-order interactions in the reduced model.

We attach one caveat. At $N = 256$ the bootstrap confidence intervals on the Sobol first-order indices remain wide — ± 0.067 for θ_P and ± 0.076 for η_{13} — so the top few entries are not separated with confidence and the second-rank magnitudes should not be over-interpreted. The broad grouping into a leading cluster, a middle band, and a negligible tail is robust; a strict ordering within the leading cluster is not.

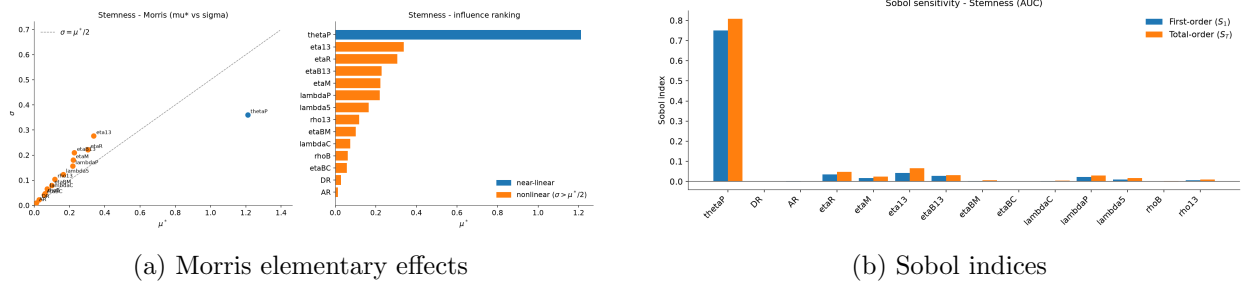


Figure 2: Sensitivity of the stemness index. The Morris panel plots μ^* against σ with the $\sigma = \mu^*/2$ nonlinearity line alongside the μ^* ranking; the Sobol panel pairs first-order and total-order indices. These panels are rendered from the original (θ_P -unclipped) sampling box and therefore show the inflated θ_P bar discussed in the text; the corrected numbers are those of Table 4.

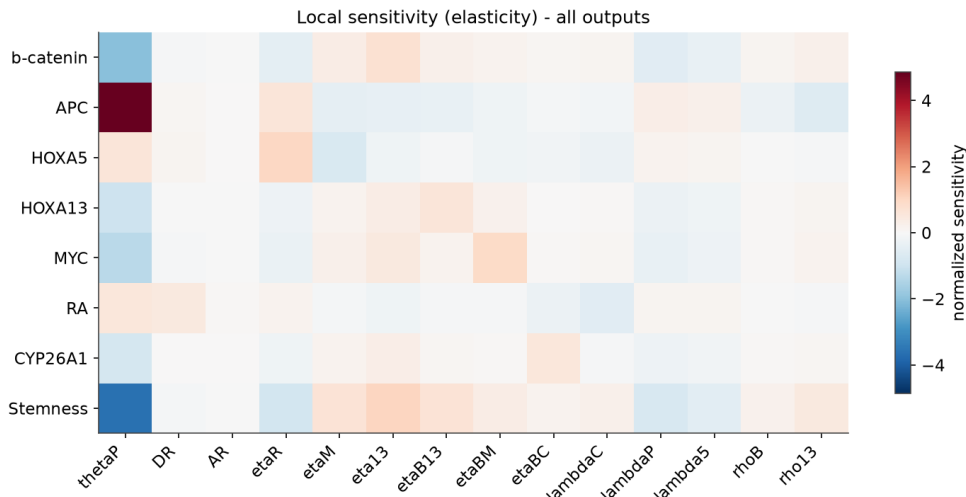


Figure 3: Signed local elasticities, eight outputs by fourteen parameters. Blue is a positive response and red negative. The θ_P column dominates and reverses sign between the tumor-suppressor outputs (APC, HOXA5) and the proliferative ones (β -catenin, HOXA13, MYC, stemness); the diagonal-like structure elsewhere is each output’s own-pathway driver. Elasticities are computed with a 5% perturbation and are unaffected by the sampling-box issue.

4.3 Comparison with the parent 27-state network

The reduced model is a lumping of a considerably larger reaction network: 27 states covering the retinoid block (including the binding proteins, receptors and their complexes), the Wnt destruction-complex block, and the HOX block, plus a derived β -catenin/TCF readout. We ran the identical three-method suite over 30 parameters of that model — 27 reaction rate constants plus three cross-pathway coupling knobs — to ask which of them the reduced model is entitled to ignore. Solves use BDF at $\text{rtol} = 10^{-3}$, $\text{atol} = 10^{-6}$, with outputs scored as time averages over a settled window, and the stemness surrogate defined analogously to Eq. (22).

The consolidation rule is deliberately conservative: a parameter is a candidate to be *fixed* only if all three methods independently score it below 5% of that method’s maximum. Anything else is *kept*. The verdict is **22 keep, 8 fix**, with the fixable set $\{v_{17}, k_{15}, k_{22}, k_{p4}, k_{m1}, k_{p9}, k_{m4}, k_{m3}\}$ and the most influential parameters being MCSynth (consensus score 0.999), v_{16} (0.830), k_{p5} (0.612), γ_1 (0.527), k_{p6} (0.464), k_{14} (0.408) and k_8 (0.351). The conservatism of the rule is visible in the near-ties: v_{17} (consensus 0.0341) is fixed while k_{23} (0.0329) is kept, because k_{23} clears the 5% bar under Morris even though its Sobol index is negligible.

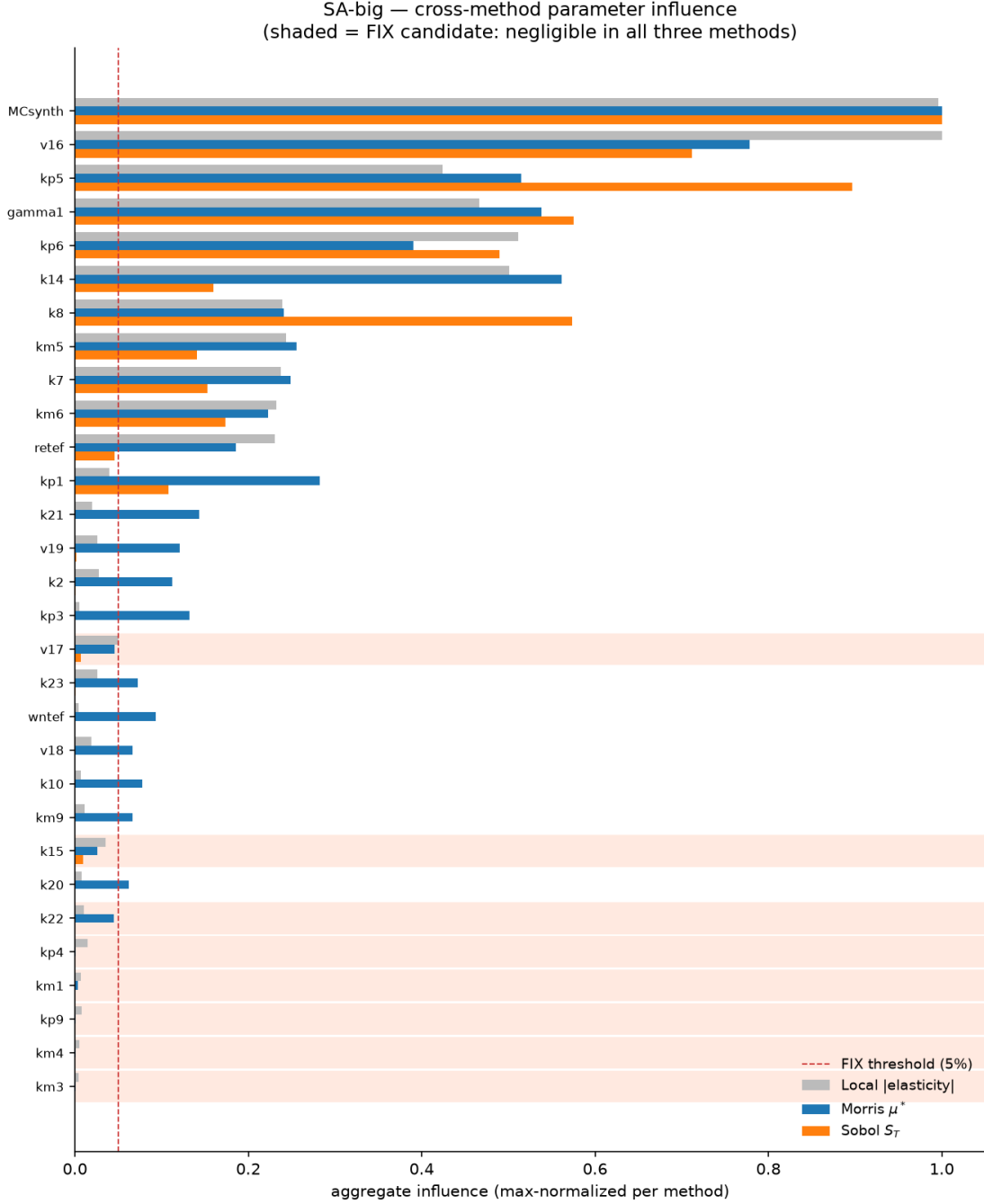


Figure 4: Cross-method consensus ranking for the 27-state model. Bars give each parameter’s max-normalized score under local elasticity, Morris μ^* , and Sobol S_T ; the dashed line is the 5% threshold and the shaded rows are the eight parameters that fall below it under all three methods.

The instructive contrast with the reduced model is in the interaction structure. Where the reduced model is nearly additive, the 27-state model is strongly interacting: $\sum_j S_T$ runs from 1.9 to 4.9 against $\sum_j S_1$ of 0.6 to 1.5. For the stemness surrogate, k_8 has $S_1 = -0.173$ but $S_T = 0.429$, and k_{p5} moves from $S_1 = 0.156$ to $S_T = 0.502$; a negative first-order index with a large total

index is the textbook signature of a parameter that acts purely through interactions. The single clean exception is γ_1 , the knob scaling β -catenin/TCF availability, which is almost purely additive ($S_1 \approx S_T \approx 0.73\text{--}0.80$ on MYC, HOXA5, and the β -catenin/TCF readout). Lumping the network down to seven states, in other words, does not merely reduce dimension — it removes most of the interaction structure, which is part of why the reduced model is a tractable inference target at all.

Three limitations of the 27-state analysis should be recorded. Roughly 7.5% of Morris solves and 8.5% of Sobol solves failed to converge and were mean-imputed. The Morris nonlinearity flag $\sigma > \mu^*/2$ fires for 27 to 30 of 30 parameters on every output, so at this sample size it is saturated and carries no discriminating information — we report it as a limitation, not a finding. And the Sobol indices for the CYP26A1 readout are numerically meaningless because its baseline settles to 7.6×10^{-6} , so the variance ratio’s denominator collapses; that row is excluded.

4.4 Scope and limitations of the sensitivity analysis

The screen establishes that the model’s outputs are strongly and interpretably sensitive to a modest number of parameters, and it identifies which. It is tempting to conclude that these are the parameters an inverse method will recover. That conclusion is wrong, and it is worth flagging here because Section 6 will demonstrate it directly: *a-priori sensitivity does not predict recoverability*. The clearest case is η_{BM} , the β -catenin→MYC arm, which has the single largest relative sensitivity of any parameter in the residual and yet recovers with a mean error of 29% and is judged non-recoverable overall. Conversely, several timescale parameters sit near the bottom of the sensitivity ranking and recover to within 10–20%. Sensitivity tells you whether a parameter can influence the data; identifiability tells you whether its influence can be *distinguished from that of the other parameters*, and it is the second question that governs recovery.

5 The forward problem: neural approximation of the model trajectories

The forward problem is to produce $y(\tau)$ on $[0, T]$ given the parameters. A stiff solver already does this to machine precision, so a PINN is not competitive as a solver here and we do not claim it is. The purpose of the forward study is different and twofold. It establishes that a network of this architecture can represent this system’s dynamics at all — a prerequisite for the inverse problem, where the same networks carry the state representation — and it isolates the one architectural choice that the system genuinely forces.

5.1 Network architecture and time embedding

The state representation is a multilayer perceptron $z_\phi : \tau \mapsto \mathbb{R}^7$ of width 256 and depth 4 with GELU activations, Xavier-normal initialization at gain 0.5, and 207,879 parameters. Time is not

fed to the network directly. It passes through a fixed random Fourier embedding [27],

$$\gamma(\tau) = \left[\frac{\tau}{T}, \sin\left(2\pi \frac{\tau}{T} B_k\right), \cos\left(2\pi \frac{\tau}{T} B_k\right) \right]_{k=1}^{16}, \quad B_k \sim \mathcal{N}(0, \sigma_F^2), \quad \sigma_F = 4, \quad (26)$$

giving an input of dimension $1 + 2 \times 16 = 33$. The frequencies B_k are drawn once and held fixed as a buffer, not trained. Retaining the raw normalized time τ/T alongside the sinusoids lets the network represent the slow trend and satisfy the initial condition without having to synthesize them from oscillatory components.

The scale σ_F is the one hyperparameter that must be matched to the problem. The circadian forcing has period $T_R = 24$ over a horizon of $T = 150$, so representing it requires a frequency near $150/24 = 6.25$ in the units of Eq. (26). The realized draw contains three frequencies that bracket this value (6.94, -6.99 , -7.16), with the remainder spread over $|B_k| \in [0.69, 7.16]$, i.e. implied periods from 21 to 216 time units. Sixteen features at $\sigma_F = 4$ therefore span both the treatment pulse and the circadian ripple, which is why this configuration suffices.

5.2 Composite loss and training protocol

We report two variants. The *dense-label baseline* fits 100 evenly spaced labeled points with loss weights $\lambda_{\text{data}} = 0.7$, $\lambda_{\text{phys}} = 0.3$ and no explicit initial-condition term; it is a supervised interpolation of a pre-solved trajectory, included as a reference point. The *sparse hybrid* is the honest forward solve and is the variant we emphasize: it sees only 40 observations drawn uniformly at random from the horizon and must reconstruct everything between them from the physics. Its objective is

$$\mathcal{L} = \lambda_{\text{data}} \mathcal{L}_d + \lambda_{\text{phys}} \mathcal{L}_p + \lambda_{\text{ic}} \mathcal{L}_{\text{ic}}, \quad \lambda_{\text{data}} = \lambda_{\text{phys}} = 1, \quad \lambda_{\text{ic}} = 20, \quad (27)$$

with

$$\mathcal{L}_d = \frac{1}{N_d} \sum_i \|z_\phi(\tau_i) - y_i\|^2, \quad \mathcal{L}_p = \frac{1}{N_c} \sum_k \left\| \dot{z}_\phi(\tau_k) - f(\tau_k, z_\phi(\tau_k); \theta) \right\|^2, \quad \mathcal{L}_{\text{ic}} = \|z_\phi(0) - y_0\|^2. \quad (28)$$

The initial condition is imposed as a weighted penalty rather than hard-wired into an ansatz. The derivative $\dot{z}_\phi$ is taken by automatic differentiation; this is unremarkable in the forward problem, where the parameters are known, but it becomes the central issue in Section 6. Collocation uses $N_c = 50,000$ points resampled uniformly at random every epoch, reduced to 10,000 during the L-BFGS phase.

The realized sparse design (seed 42) places observations with a median gap of 2.76 and a maximum gap of 13.92 time units, giving about 6.4 observations per circadian period, with 13 of the 40 falling inside the treatment window. This is a genuinely sparse and irregular design: several circadian cycles are covered by fewer than two points, so the oscillation between them is supplied by the physics residual, not by interpolation.

Training is two-stage: Adam [34] for 1000 epochs at learning rate 10^{-3} with cosine annealing

to 10^{-6} , followed by 500 outer steps of L-BFGS [35] with strong-Wolfe line search, history size 50 and up to 20 inner iterations per step. All computation is in float64.

5.3 Accuracy against the reference solution

Table 5 reports the error of the trained networks against the Radau reference on a uniform grid of 6000 points spanning the full horizon. We give both relative L^2 error and NRMSE (RMSE normalized by the range of the reference), for a reason visible in the table: APC settles to a very small value in the high-WNT regimes, so its relative L^2 error is inflated by a vanishing denominator. In Severe APC Loss the APC relative L^2 is 13.53% while its NRMSE is 1.26%, entirely in line with every other state. NRMSE is the fair summary for that species.

Regime	β -catenin	APC	HOXA5	HOXA13	MYC	RA	CYP26A1	mean	NRMSE
<i>Sparse hybrid, 40 random observations + IC + physics</i>									
Normal	2.93	5.85	1.79	2.93	2.69	1.71	1.93	2.83	2.86
Early Adenoma	1.56	2.84	1.61	1.80	1.59	1.74	1.16	1.76	1.51
Advanced Adenoma	0.91	7.02	1.69	1.33	1.02	1.75	0.89	2.09	1.20
Severe APC Loss	0.86	13.53	1.94	1.18	0.86	1.49	0.81	2.95	1.16
<i>Dense-label baseline, 100 evenly spaced labels</i>									
Normal	0.39	0.74	0.70	0.21	0.25	0.73	0.38	0.49	0.46
Early Adenoma	0.34	1.27	0.95	0.25	0.41	1.00	0.34	0.65	0.49
Advanced Adenoma	0.27	4.14	1.43	0.36	0.56	1.06	0.37	1.17	0.62
Severe APC Loss	0.36	9.11	1.41	0.34	0.55	1.26	0.37	1.91	0.68

Table 5: Forward PINN accuracy: per-state relative L^2 error (%) against the Radau reference over $\tau \in [0, 150]$, with the per-regime mean and mean NRMSE (%) in the last two columns. Grand means: sparse hybrid 2.41% relative L^2 (1.59% excluding APC), 1.68% NRMSE; dense baseline 1.06% relative L^2 , 0.56% NRMSE.

The headline comparison is the one the two halves of the table invite. The sparse hybrid reaches 2.41% mean relative L^2 from 40 randomly placed observations, against 1.06% for the dense baseline from 100 evenly spaced ones. Reconstructing the full seven-state trajectory to within roughly $2.4\times$ the dense-label error, from $2.5\times$ fewer and irregularly spaced points, is what the physics term buys: the residual constrains the solution everywhere on the horizon, including across the wide gaps where no data exist. We reproduced this comparison in two independent runs a month apart with essentially identical results (sparse-hybrid per-regime means of 2.90/1.71/2.24/2.96 in the earlier run), so the figures are stable under rerun, though every run uses the same seed and we therefore have no estimate of seed-to-seed variability.

The composition of the final loss is worth noting as a diagnostic. In the sparse hybrid the physics term accounts for 94–97% of the total loss at the end of Adam training; the data term is essentially satisfied early and the optimizer spends the remainder of its budget enforcing the ODE. In the dense baseline the two terms remain the same order of magnitude throughout. This is the expected signature of a genuinely physics-driven solve versus a data-driven one.

Core WNT-HOX Regulatory Dynamics via Forward PINN (40 sparse observations)

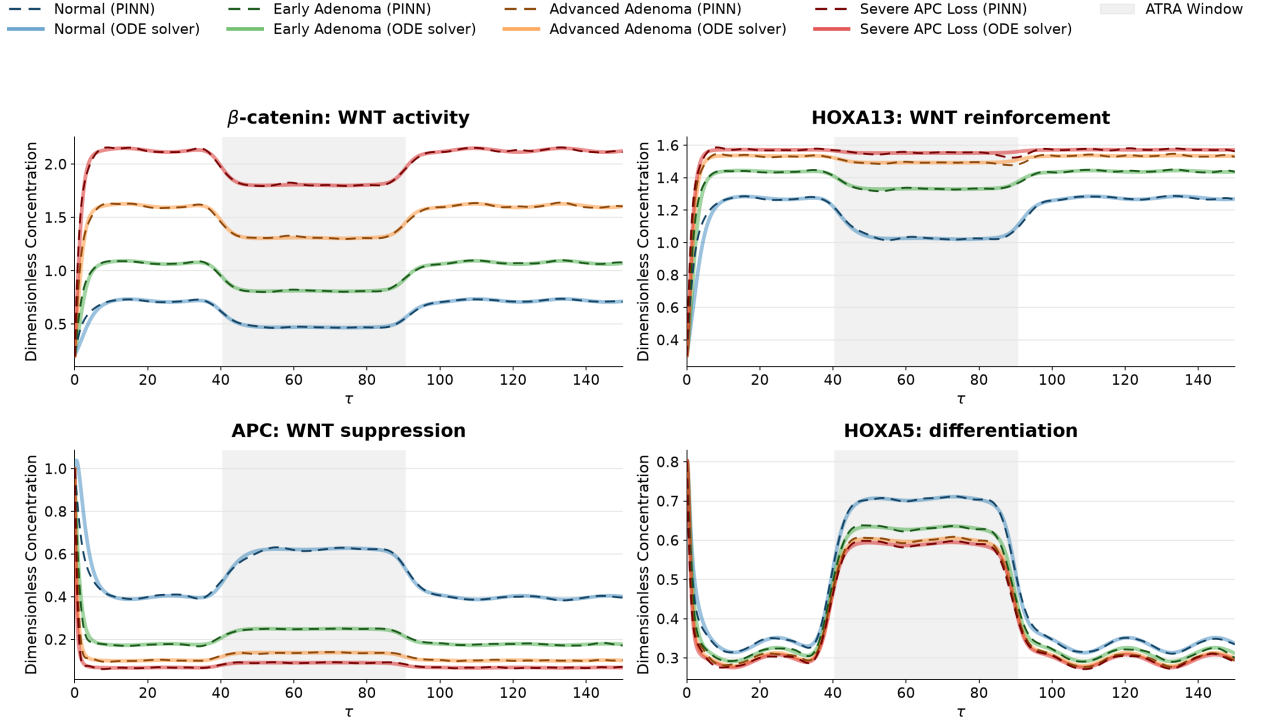


Figure 5: Forward PINN reconstruction of the four core WNT–HOX species across all four regimes, from the *sparse hybrid* model — 40 random observations plus the physics residual. Thick translucent curves are the **Radau** reference and thin dashed curves the PINN. They are visually indistinguishable over almost the whole horizon; the one place the discrepancy is apparent to the eye is the steep initial transient ($\tau \lesssim 5$), where APC falls from 1.0 and HOXA13 rises from 0.3 faster than the network tracks. That transient is the largest contributor to the Normal regime’s error in Table 5, and it is where the observation design is thinnest. The shaded band is the *half-response window* — the median 50% crossing of the switch-on and switch-off transitions across states and regimes — which sits at $[40.4, 90.7]$, lagged relative to the dose window $[40, 88]$ because the states respond with delay.

5.4 Spectral bias and the necessity of the Fourier embedding

The Fourier embedding is not a refinement; without it the model does not solve the problem. A plain MLP taking τ directly as a scalar input (199,687 parameters, otherwise identically trained) plateaus early and never recovers. Its data loss stalls at 1.41×10^{-3} and its physics loss at 1.83×10^{-2} by around epoch 2000 and does not improve thereafter; 500 subsequent L-BFGS steps move the total loss from 6.48×10^{-3} to 6.54×10^{-3} , that is, not at all. The Fourier-embedded network reaches a data loss of 1.9×10^{-5} , roughly 75 times lower.

The failure is specifically a failure to represent the fast component, exactly as spectral-bias theory predicts [24, 25]. In the Normal regime the plain network’s RA output spans $[0.161, 0.440]$ where the true RA spans $[0.156, 0.787]$: it captures under half the amplitude of the ATRA pulse and none of the circadian ripple, having settled on a smooth curve through the middle of the

oscillation. The Fourier-embedded network on the same regime spans $[0.1563, 0.7868]$ against a true $[0.1562, 0.7867]$, agreeing to three or four decimal places. Restricted to the active window, the plain network’s mean relative L^2 error is 23–26% across the four regimes.

We must disclose a confound: the surviving plain-MLP run was trained over a horizon of $T = 3000$ rather than $T = 150$, so the embedding and the horizon differ simultaneously and this is not a controlled ablation. The direction and magnitude of the effect are unambiguous and consistent with the established theory, but a clean single-variable ablation at fixed T was not run and we do not claim one.

5.5 Summary of the forward results

The forward study supports a narrow but necessary claim. A Fourier-feature network of this size represents the seven-state system across all four regimes to roughly 1–3% error, and does so from 40 sparse observations by leaning on the physics residual. The multi-timescale forcing makes the time embedding mandatory rather than optional. Both facts carry directly into the inverse problem, where the same architecture supplies the state representation and the question shifts from whether the trajectory can be represented to whether the parameters that generated it can be recovered.

6 The inverse problem: parameter estimation from trajectory data

6.1 Problem formulation and scoring criterion

The inverse problem is to recover $\theta \in \mathbb{R}^{36}$ from sparse noisy observations of the trajectory. In the PINN formulation the parameters become trainable variables optimized jointly with the state networks [16], and the objective is Eq. (27) with θ now free. Three design decisions apply throughout.

Multiple conditions, shared parameters. We train one state network per experimental condition but a single shared parameter vector, so θ must satisfy the residual under all conditions simultaneously. This is the mechanism by which additional experiments constrain the parameters rather than merely adding degrees of freedom.

Log parameterization. Every positive parameter is represented as $\theta_j = \theta_j^{\text{nom}} \exp(s_j)$ with s_j trainable, and θ_P as $\text{sigmoid}(s)$ mapped into $(0, 1)$. This enforces positivity by construction and, more importantly, makes the gradient with respect to s_j a *relative* sensitivity, so parameters whose true values differ by two orders of magnitude receive comparably scaled updates.

Honest initialization. All estimates start at $1.5\times$ the true value, with W and θ_P additionally displaced to deliberately wrong values. No optimization is ever started at or near the truth.

We score recovery as the number of the 36 parameters whose estimate falls within 10% relative error, reported per regime. This threshold is a convention, not a claim about what precision biology requires; we use it consistently so that comparisons between methods are meaningful, and we report

per-regime counts rather than a pooled figure because the regimes differ sharply in difficulty.

6.2 A two-parameter proof of concept

We first restricted the unknowns to (W, θ_P) , the two parameters that define the regimes. From a single condition with 80 observations at 1% noise, the result already contains the entire subsequent story in miniature (Table 6).

Regime	W true	W error	θ_P true	θ_P error
Normal	0.80	1.15%	1.00	0.63%
Early Adenoma	1.00	0.39%	0.75	1.12%
Advanced Adenoma	1.50	2.87%	0.50	23.5%
Severe APC Loss	2.00	5.12%	0.25	142.0%

Table 6: Two-parameter recovery from a single condition. W recovers to within a few percent everywhere. θ_P recovers cleanly in the low-WNT regimes and fails progressively as WNT drive increases; in Severe APC Loss it barely leaves its (deliberately wrong) initialization of 0.60.

With only two unknowns and abundant data, the problem is as favorable as it will ever be, and θ_P still fails in the high-WNT regimes. This is the first appearance of the wall analyzed in Section 7, and it establishes early that the difficulty is not one of scale.

Extending the same setup to all 36 parameters, still from a single condition, gives 4/8/7/7 parameters under 10% and a mean relative error of 47%. One condition does not constrain 36 parameters, which is unsurprising; the rest of this section is about what does.

6.3 Multiple experimental conditions and improved conditioning

We next added experimental conditions — variations in the ATRA protocol ($D_R = 0$; early, late, low-dose windows; a strengthened circadian amplitude) — along with the log parameterization, decoupled learning-rate schedules for network weights and parameters, and adaptive gradient-norm loss weighting [23, 36]. The decoupled schedules fixed a real defect: a single cosine annealing schedule had been driving both the network weights and the physical parameters to learning rates of order 10^{-6} , freezing the parameters partway through training.

Going from 1 to 3 to 6 conditions moved the count from 26 to 30 to 26 across the four regimes, with the 6-condition PINN landing at **10/4/5/7**. The improvement is real but small, and it does not scale with the number of conditions.

The reason is that all of these conditions perturb the same input. Every one of them modulates the RA forcing $\mu_R(\tau)$, and RA reaches the WNT/HOX/MYC subnetwork through exactly one door: the term $\eta_R r / (\kappa_R + r)$ in the h_5 equation. Adding a fourth or a seventh way of shaping the same input does not illuminate parameters that lie behind that door. A controlled test of this makes the point precisely: holding everything else fixed and increasing the number of RA-protocol conditions from 3 to 6 to 8 moves the recovery count $5 \rightarrow 5 \rightarrow 7$ of 36 while the error on W gets

monotonically *worse* (35.1% $\rightarrow$ 38.1% $\rightarrow$ 39.0%). More experiments of the same kind are close to worthless. This observation motivates both remaining levers.

6.4 A negative result: network capacity is not the limiting factor

Before pursuing those levers we tested the most obvious hypothesis — that a network with 207,879 parameters per condition is simply too flexible, and absorbs the residual at incorrect θ . We shrank the network from 4×256 to 3×128 (a $5.4\times$ reduction, to 38,279 parameters), added weight decay of 10^{-4} applied to the network only and never to the physical parameters, and replaced joint optimization with an alternating two-timescale scheme (five network-only steps, then five parameter-only steps on the physics residual with the networks frozen).

The result was **9/10/2/4**, a total of 25 against the 26 of the unmodified 6-condition run. Both levers together made recovery marginally worse. We report this because it is genuinely informative: capacity control is the intuitive remedy and it does not work, which redirects attention to the estimator and to the data.

One diagnostic from this run deserves to be quoted, because it states the underlying pathology more sharply than any aggregate count. During the frozen-network parameter refinement on the Normal regime, W sat pinned at 0.613 against a true value of 0.80 while the physics loss remained flat at 4.5×10^{-5} for all 500 steps. The optimizer was at a minimum of the residual. The minimum was simply not at the true parameters.

6.5 Diagnosis: gradient starvation and derivative bias

Two distinct mechanisms produce that outcome, and separating them is what makes the rest of the section possible.

Gradient starvation. The 36 parameters enter the objective *only* through the physics residual, whereas the network weights enter through every term. With 623,637 network weights driving the residual down, the residual can be made small at incorrect θ , after which the gradient with respect to θ all but vanishes. The signature is direct: in the 3-condition run on the Normal regime, W reached 0.974 by Adam epoch 100 and then sat at 0.989–0.990 from epoch 500 through 1800 — effectively frozen — while the total loss fell by a factor of 360 and the physics loss by a factor of 178. The optimizer made three orders of magnitude of progress on the objective and none on the parameter. Increasing the data by $5\times$ and reducing the noise by $5\times$ changes nothing: W still settles at 0.989.

Derivative bias. The second mechanism is subtler and, it turns out, the more consequential. The residual $\dot{z}_\phi - f(z_\phi; \theta)$ requires $\dot{z}_\phi$ by automatic differentiation. A network can fit a trajectory accurately while its derivative is noticeably wrong — accuracy in z does not imply accuracy in $\dot{z}$ — and any such error enters the residual as a bias that θ then compensates for.

We isolated this with a controlled experiment. Networks were first fitted to data alone, reaching trajectory RMSE of 0.0045–0.0055 against the reference. The networks were then frozen and the parameters alone were optimized against the physics residual. The residual fell from 3.70×10^{-1} to 2.39×10^{-4} in 50 steps and then stayed exactly flat for 250 more. At that minimum, W landed at 0.504 against a true 0.80 — a 37% error, *worse* than the 0.98 obtained in the joint run despite a residual 1550 $\times$ lower — and only 6 of 36 parameters fell within 10%.

That is the derivative-bias ceiling in a single number. Lower residual, worse parameters. The minimum of the residual in θ -space is not at the true θ , because the residual is evaluated against a derivative that is systematically slightly wrong. No amount of better optimization fixes this; the objective itself is misplaced.

6.6 A derivative-free integral residual

If the biased $\dot{z}_\phi$ is the problem, the remedy is an equivalent residual that never forms it. Integrating the ODE between adjacent collocation points and applying the trapezoidal rule gives

$$r_k = (z_{k+1} - z_k) - \frac{1}{2} \Delta \tau_k (f_k + f_{k+1}), \quad f_k \equiv f(\tau_k, z_\phi(\tau_k); \theta), \quad (29)$$

on a fixed sorted grid. This is a multiple-shooting-style constraint that uses *network values only* and never differentiates the network with respect to time. It is accompanied by three supporting changes: relative-error weighting $w = 1/\text{scale}$ applied to both the data and residual terms so that species with different dynamic ranges contribute comparably; deterministic collocation on a fixed uniform grid rather than fresh random points each step, which Eq. (29) requires; and two restarts, keeping the run with the lower final physics residual.

The result across the four regimes is **17/16/10/7**, a total of 50. Against the 6-condition derivative-residual baseline of **10/4/5/7** this is +24, but that comparison is not clean — the baseline used 6 conditions and the integral run used 10 — so it conflates this lever with the next one. The controlled comparison holds the conditions fixed at 10 and changes only the residual: **37** $\rightarrow$ **50, a gain of 13 parameters**, and that is the figure we stand behind for this lever.

Inverse PINN — true vs recovered (8 best-converging params)

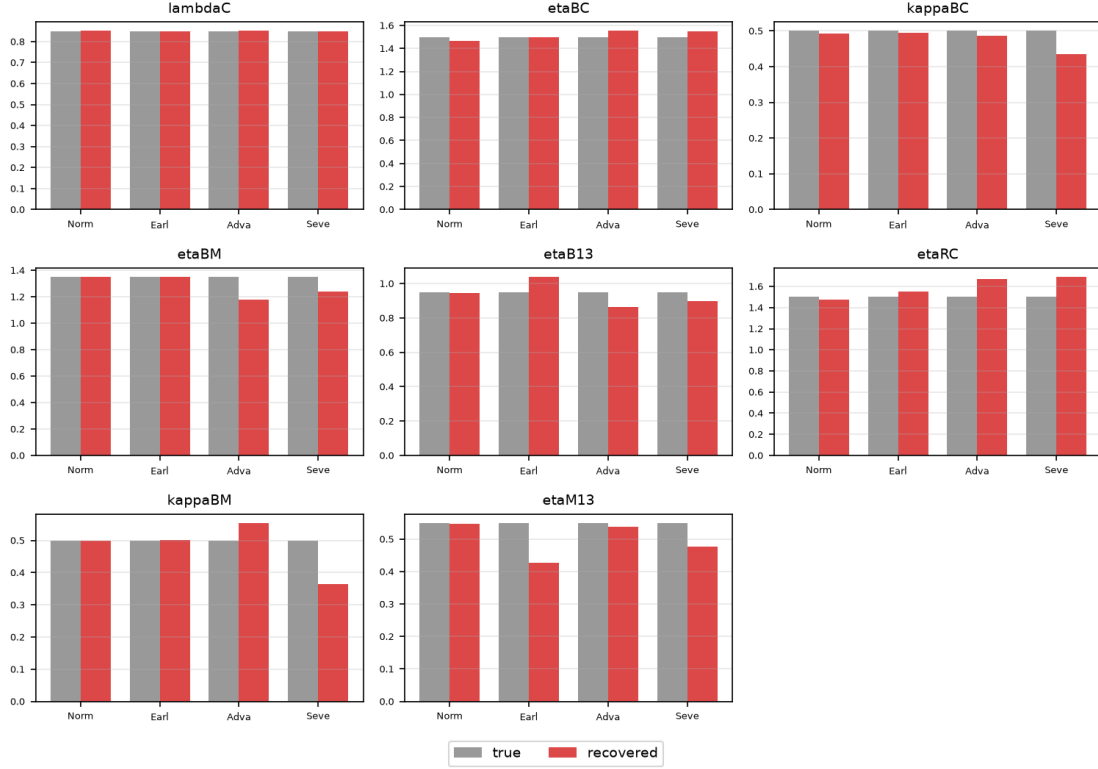


Figure 6: Recovered versus true values for the eight best-recovered parameters under the integral-residual PINN, across the four regimes.

Two caveats belong with this result. First, the restart-selection rule — keep the run with the lower final physics residual — picked the *worse*-counting restart in 3 of the 4 regimes. Selecting the better restart in each case would have given $17/17/13/8 = 55$ rather than 50. Residual value and recovery count are anti-correlated here, which is the same phenomenon the diagnosis above describes, now visible in the model-selection step: the residual is not a reliable proxy for parameter accuracy even when comparing two fits of the same model. Second, the planned ablation separating the integral residual from the relative weighting, fixed collocation, and restarts did not complete — two of its four arms crashed. The +13 is therefore attributable to the *bundle* of four changes, and we cannot apportion it among them. In particular we make no claim about the integral residual in isolation, and the SIREN variant [37] was never evaluated.

6.7 Parameter-exciting experimental conditions

The second lever follows from the observation that all our conditions perturbed the same input. We added two exogenous input channels that the base model does not use — a direct drive D_W into the β -catenin equation and D_M into the MYC equation — and defined four new conditions with them: a WNT pulse ($D_W = 1.0$ over $\tau \in [30, 70]$), which sweeps β -catenin across the Hill half-

saturation scales κ_B ; a sustained WNT inhibition step ($D_W = -0.7$ from $\tau = 50$), aimed directly at de-saturating β -catenin in the high-WNT regimes; a MYC pulse; and a combined WNT/RA condition. These are known, experimenter-controlled inputs, held fixed and not estimated, and they are zero by default so the base model is unchanged.

Scored with the classical ODE-fit estimator (Section 6.8), moving from 6 RA-protocol conditions to these 10 lifts recovery from **18/17/13/13** (total 61) to **24/23/21/14** (total 82), a gain of 21 parameters. In the Normal regime θ_P recovers to 0.9% and in Advanced Adenoma W recovers to 0.0% — the de-saturating condition does exactly what it was designed to do. Fourteen parameters now recover to within 10% in *all four* regimes: λ_C , η_{RC} , κ_{RC} , κ_R , κ_{B13} , κ_{BC} , κ_{BM} , η_{B13} , η_{BM} , ε_{13} , ε_C , ε_M , ε_R , ρ_5 .

This is the largest single gain reported in this paper, and it comes from changing the *kind* of information rather than its quantity. Recall that increasing the amount of same-kind information moved the count $5 \rightarrow 5 \rightarrow 7$. Four conditions that open new doors into the network are worth more than any number that open the same one.

Even so, the wall holds where it matters most. In Severe APC Loss the count reaches only 14/36 and θ_P is off by 264%.

6.8 Comparison with classical ODE-constrained optimization

Throughout we compare the PINN against a classical estimator on identical data: integrate the ODE with a stiff solver under each condition, stack the residuals against the same sparse noisy observations, and solve the nonlinear least squares problem in the same log/sigmoid coordinates with a trust-region method, from four jittered restarts. This is standard multiple-condition shooting, and it shares neither of the PINN’s two defects — there is no network to absorb the residual, and the derivative is supplied by the integrator, not by differentiating a fit.

Table 7 collects the whole progression.

Method	Key change	cond.	Norm.	Early	Adv.	Severe	total
PINN, single condition	36 unknowns	1	4	8	7	7	26
PINN, multi-condition	log-param, adaptive λ	3	8	12	6	4	30
PINN, multi-condition	+ Stage-3 refine	6	10	4	5	7	26
PINN, smaller nets	3×128 , alternating opt.	6	9	10	2	4	25
PINN, exciting cond.	WNT/MYC channels	10	16	9	6	6	37
PINN, integral residual	derivative-free	10	17	16	10	7	50
ODE-fit	multiple shooting	6	18	17	13	13	61
ODE-fit, exciting cond.	+ relative weighting, 4 restarts	10	24	23	21	14	82

Table 7: Parameters recovered within 10% relative error, out of 36, per regime. Every row uses the same model, the same data recipe, the same seed, and an initialization displaced from the truth.

The table supports three conclusions.

First, **the PINN’s ceiling is lower than the classical estimator’s, and the gap is the derivative**. Every PINN variant that retained the autodiff residual stayed in the 25–37 band

regardless of conditions, architecture, weighting, or optimization schedule. The classical estimator, which never forms that derivative, was at 61 with fewer conditions. Removing the derivative from the PINN closed roughly half the gap, taking it to 50.

Second, **information is the only lever that moves both estimators**. It took the ODE-fit from 61 to 82 and the PINN from 26 to 37 with the residual held fixed. Neither architecture nor optimization did anything comparable.

Third, **the ordering across regimes is invariant**. Every row degrades from Normal to Severe APC Loss. No method reverses it. That invariance is what distinguishes a property of the problem from a property of the estimator, and it is the subject of Section 7.

6.9 Which parameters are recovered

Aggregating across 28 attempts (seven major runs times four regimes) gives a per-parameter reliability that is far more stable than any single run, and that has clear structure (Table 8).

Reliability	Parameters
Always (28/28)	λ_C
Usually ($\geq 16/28$)	η_{BC} (22), η_{RC} (19), κ_{BC} (19), η_{BM} (16)
Often (10–13/28)	η_{B13} , κ_{BM} , ε_{13} , ε_M , κ_{13} , ε_C
Sometimes (7–9/28)	η_{13} , κ_{B13} , η_{M13} , W (8), ρ_5 , κ_R , κ_{RC} , ε_R
Rarely (4–6/28)	ρ_B , ρ_{13} , ε_5 , a_5 , θ_P (5), δ_{P1} (5), η_R , η_M , κ_{M13} , a_M , λ_P , a_C
Essentially never	λ_5 (3), ε_P (3), a_{13} (3), κ_5 (1), κ_M (0)

Table 8: Per-parameter recovery reliability: the number of times out of 28 attempts each parameter fell within 10% relative error.

The failures are structured rather than random, and they fall into three groups. The largest consists of **correlated η/κ Hill pairs** — (η_R, κ_R) , (η_M, κ_M) , $(\eta_{M13}, \kappa_{M13})$ — in which the maximum induction rate and the half-saturation constant compensate for one another: one overshoots, its partner absorbs the discrepancy, and the trajectory is unchanged. The second consists of **basal production rates** (a_5 , a_M , a_{13} , a_C), which are additive constants competing with whatever else adds to the same equation. The third is δ_{P1} together with θ_P , discussed below. Notably, the reliably recovered parameters λ_C , η_{BC} , η_{RC} , η_{BM} , κ_{BC} are precisely those on the arms that the exciting conditions drive directly.

Finally, the negative result promised in Section 4: **a-priori sensitivity does not predict recoverability**. The parameter with the single highest relative sensitivity in the residual, η_{BM} , recovers with a mean error of 20% and per-regime errors of 0.47/1.47/29.4/49.0%, and is judged non-recoverable overall. Meanwhile ε_5 , ε_{13} and ε_M sit at half a percent of the maximum sensitivity and recover to 10–20%. High sensitivity means a parameter moves the data; it says nothing about whether some *other* parameter can move the data the same way. Only an identifiability analysis answers that.

7 Identifiability analysis

Section 6 showed a recovery ceiling that no estimator crossed and a regime ordering that no estimator reversed. This section asks whether that is a property of the problem rather than of our methods, using three independent tools: an analytic argument, the Fisher information matrix, and profile likelihood. A fourth view, the Bayesian posterior, follows in Section 8.

A terminological note first. *Structural* identifiability asks whether distinct parameter vectors can produce identical model output for all time, given perfect noise-free data; it is a property of the equations alone [14, 15]. *Practical* identifiability asks whether the data actually available, at the noise level actually present, can distinguish them [38, 39]. We did not run a symbolic structural identifiability analysis, so with one exception noted below every claim in this section is a practical, conditioning-based one.

7.1 An analytically exact degeneracy

The exception is a degeneracy visible by inspection. APC functionality and the excess-degradation coefficient enter the model only through

$$\delta_P(\theta_P) = 1 + \delta_{P1}(1 - \theta_P), \quad (30)$$

i.e. only through the *product* $\delta_{P1}(1 - \theta_P)$. Any pair (δ_{P1}, θ_P) preserving that product produces an identical trajectory, so the two parameters are structurally unidentifiable as a pair and only their combination is determined. Two consequences follow immediately. In the Normal regime $\theta_P = 1$ exactly, so δ_{P1} vanishes from the model and is completely undetermined — no data can constrain it. And in every other regime θ_P can only ever be recovered as accurately as δ_{P1} is known, and vice versa.

This is a modest observation, but it makes a useful calibration check: any identifiability diagnostic worth trusting should find it without being told. All three of ours do.

7.2 Fisher information spectra

We compute the Fisher information matrix in log-parameter coordinates,

$$S_{ij} = \theta_j \frac{\partial y_i}{\partial \theta_j}, \quad F = \frac{1}{\sigma^2} S^\top S, \quad \sigma = 0.002, \quad (31)$$

with sensitivities obtained by central differences at the true parameters of each regime. All seven states are treated as observed under all ten excitation conditions on a uniform grid of 200 time points, giving 14,000 observation rows against 36 parameters — a nominally very over-determined system.

The eigenstructure of F is the object of interest. Its inverse is the Cramér–Rao bound on the covariance of any unbiased estimator, so an eigenvalue $\lambda \ll 1$ corresponds to a direction

in parameter space along which the data cannot constrain the parameters to better than 100% relative uncertainty. We deliberately avoid reading identifiability off the pseudo-inverse diagonal: at condition numbers of 10^{17} a pseudo-inverse silently truncates exactly the directions we are trying to measure. Instead we count directions with $\lambda < 1$ (hard-null) and $\lambda < 10^{-4}\lambda_{\max}$ (sloppy), and assign each parameter a verdict from its participation $(\sum_{k \in \text{null}} V_{jk}^2)^{1/2}$ in those directions.

Regime	$\text{cond}(F)$	null	sloppy	I/W/N	$ R > 0.99$	non-identifiable
Normal	3.7×10^{19}	1	17	13/22/1	0	δ_{P1}
Early Adenoma	7.9×10^{13}	1	14	17/18/1	1	δ_{P1}
Advanced Adenoma	1.9×10^7	1	18	14/20/2	6	θ_P, δ_{P1}
Severe APC Loss	1.8×10^{14}	1	19	14/21/1	10	θ_P

Table 9: Fisher information analysis of the full 36-parameter problem. “null” counts eigendirections with $\lambda < 1$; “sloppy” counts $\lambda < 10^{-4}\lambda_{\max}$; I/W/N are the identifiable, weak and non-identifiable verdicts; $|R| > 0.99$ counts parameter pairs whose posterior correlation exceeds 0.99.

Four readings of Table 9 matter.

Every regime has exactly one hard-null direction. Not zero, and not a number that varies with regime. That is the analytic (θ_P, δ_{P1}) degeneracy, present in all four regimes because it is a property of the equations.

The diagnostic finds the analytic degeneracy unprompted. The correlation between θ_P and δ_{P1} is 1.0000 in Advanced Adenoma and Early Adenoma and 0.9996 in Severe APC Loss. In Normal, where $\theta_P = 1$ removes δ_{P1} from the model entirely, δ_{P1} has null participation 1.0000 — completely unconstrained — while θ_P itself is judged identifiable. The numerics reproduce the algebra exactly.

The number of degenerate pairs grows monotonically with WNT drive: $0 \rightarrow 1 \rightarrow 6 \rightarrow 10$. This is the quantitative form of the wall. In Severe APC Loss the correlated pairs are (a_M, η_{BM}) at -0.9999 , (θ_P, δ_{P1}) at $+0.9996$, (η_R, η_M) at $+0.9994$, $(\eta_{B13}, \kappa_{B13})$ at -0.9989 , (a_C, η_{BC}) at -0.9985 , $(\varepsilon_P, \rho_{13})$ at -0.9983 , (W, η_{13}) at -0.9975 , and three more. Every one is a compensation of the form identified empirically in Section 6: a basal rate against a maximum induction rate, or a maximum induction rate against its own Hill constant. The FIM and the recovery statistics are describing the same structure from opposite directions.

The spectrum is sloppy in the standard sense. Fourteen to nineteen of 36 directions fall below 10^{-4} of the leading eigenvalue, and the spectrum in the Normal regime runs from 7.9×10^9 down to 2.1×10^{-10} . This is the characteristic pattern of systems-biology models [40, 41]: a few stiff directions that the data pin down tightly and a long tail of soft directions that they barely constrain at all. Our model is not unusual in this respect, which is part of the point — the difficulty is generic.

We note that the condition number is not monotone in WNT drive (Advanced Adenoma is the smallest at 1.9×10^7) and should not be read as a difficulty ranking. The count of near-degenerate pairs is the better-behaved statistic, and it is monotone.

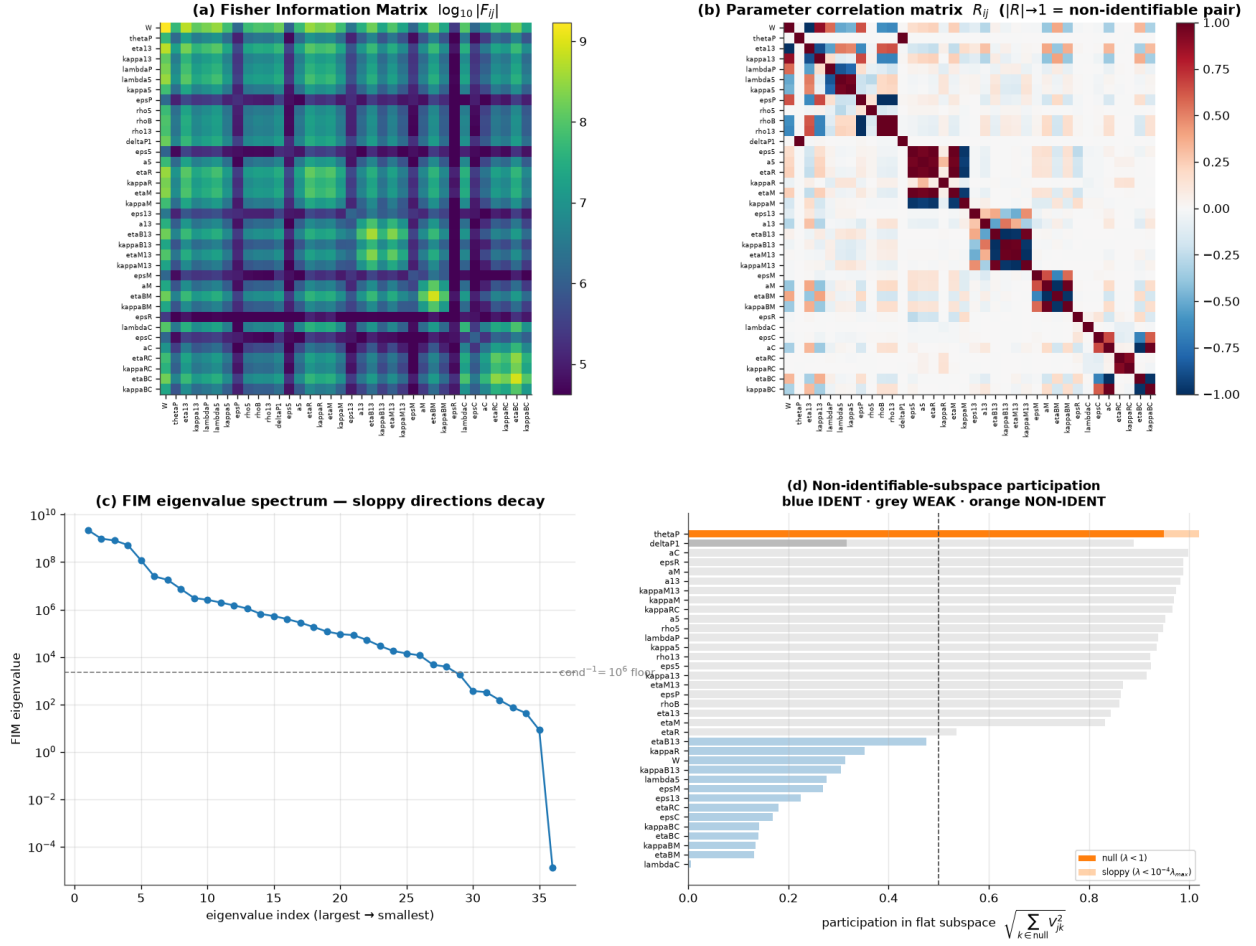


Figure 7: Fisher analysis of the Severe APC Loss regime, full 36-parameter problem: (a) magnitude of the information matrix, (b) the parameter correlation matrix, (c) the eigenvalue spectrum on a logarithmic axis, and (d) each parameter’s participation in the flat subspace, colored by verdict. The long tail in panel (c) and the off-diagonal blocks in panel (b) are the sloppiness and the compensating pairs respectively.

7.3 A well-posed reduced parameter set

If the difficulty is a property of the 36-parameter problem rather than of the model or the data, then restricting to the parameters the data actually constrain should give a well-posed problem. We tested this by selecting the eight parameters with the lowest flat-subspace participation across all regimes — λ_C , η_{BM} , κ_{BM} , κ_{BC} , η_{BC} , η_{RC} , κ_{B13} , κ_R — fixing the other 28 at their true values, and recomputing the Fisher matrix on identical data.

Regime	$\text{cond}(F)$	null	sloppy	verdict
Normal	99.7	0	0	8/8 identifiable
Early Adenoma	120.9	0	0	8/8 identifiable
Advanced Adenoma	229.4	0	0	8/8 identifiable
Severe APC Loss	665.2	0	0	8/8 identifiable

Table 10: The eight-parameter subproblem on identical data. The condition number falls from 10^7 – 10^{19} to 10^2 – 10^3 , with no null directions, no sloppy directions, and no correlated pairs in any regime.

The contrast is stark: fifteen to seventeen orders of magnitude of conditioning recovered by removing 28 parameters. It also shows that the data are not impoverished in any absolute sense — they contain enough information to determine eight parameters comfortably in every regime, including the worst. The problem is that we were asking them to determine 36. Note also that the residual regime ordering survives even here: the condition number still climbs from 99.7 to 665.2 across the progression. The wall is attenuated, not eliminated, by reducing the parameter set.

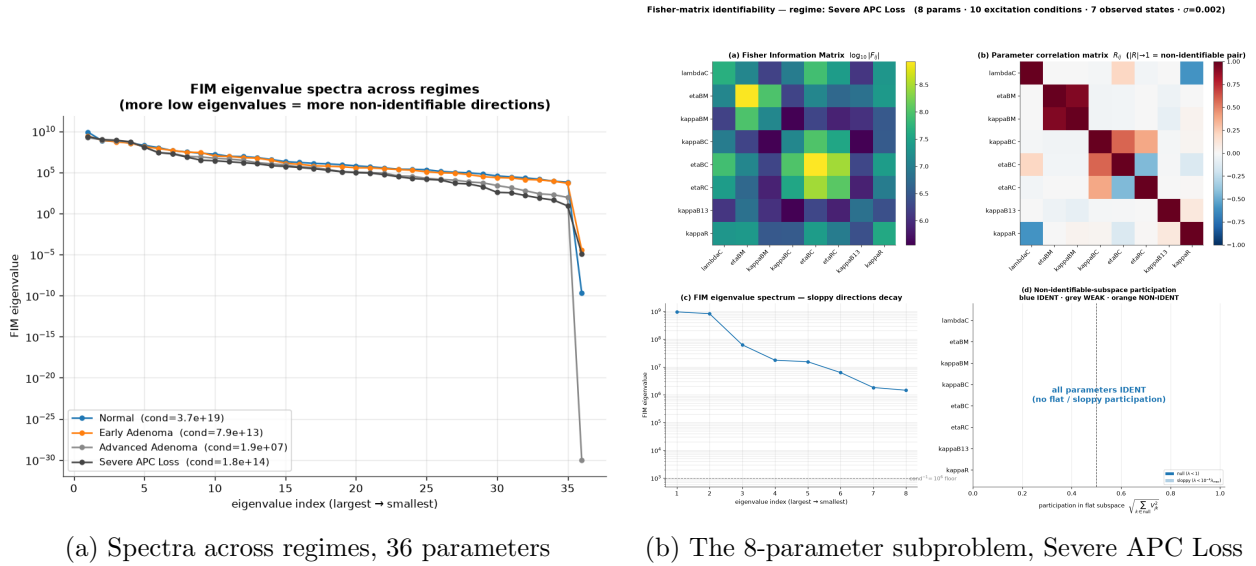


Figure 8: Left: eigenvalue spectra of the full 36-parameter information matrix for all four regimes on a common logarithmic axis, showing the long soft tail in each. Right: the same analysis restricted to the eight best-constrained parameters in the worst regime — a compact spectrum, no flat directions, and all eight parameters identifiable.

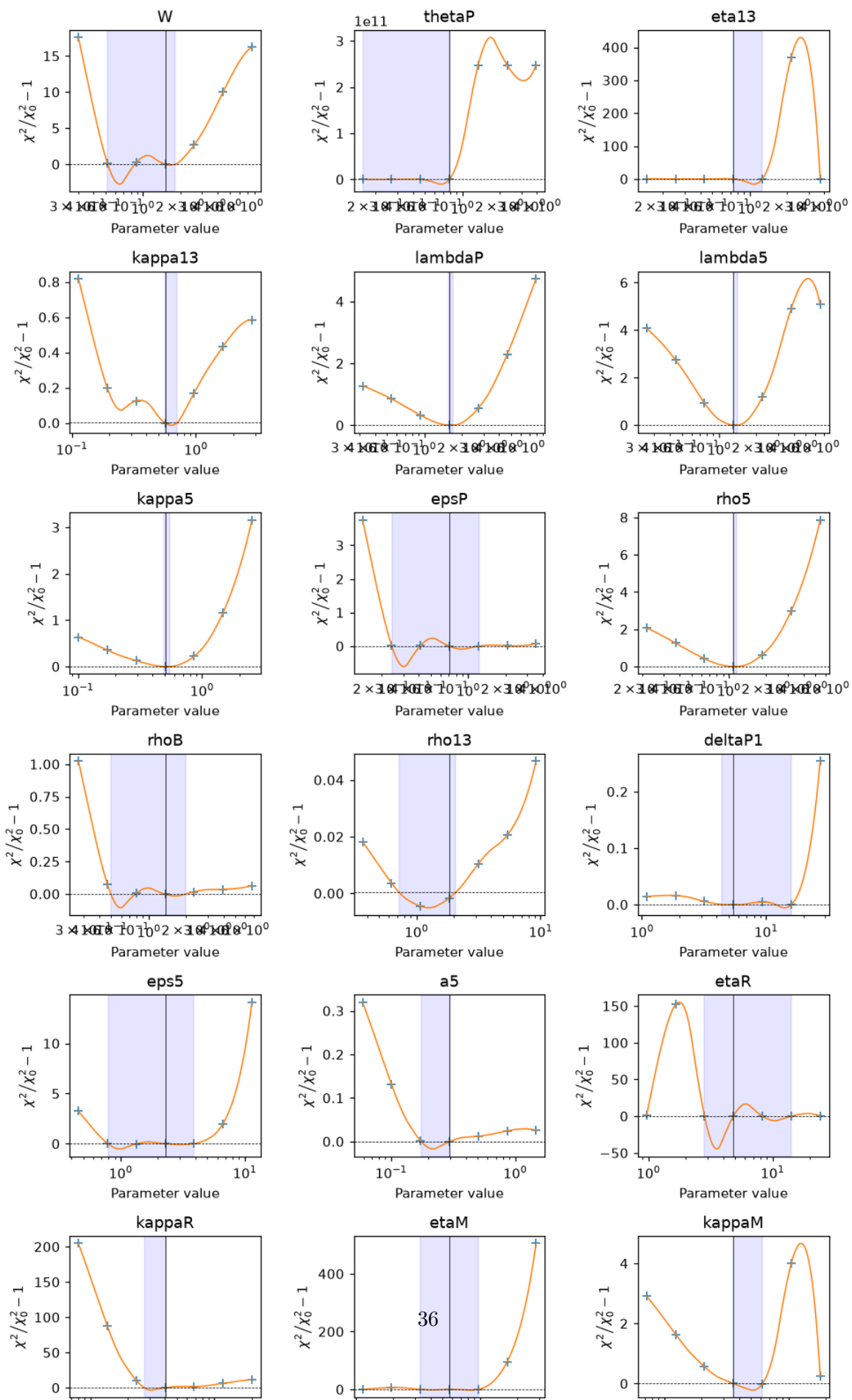
7.4 Profile-likelihood confidence intervals

The Fisher matrix is a local, quadratic approximation evaluated at the true parameters. Profile likelihood [38] makes no such approximation: each parameter is scanned over a wide range while all others are re-optimized at every point, and the confidence interval is read off where the profile crosses a χ^2 threshold. It is the more trustworthy of the two and far more expensive.

Starting from the exciting-conditions ODE-fit optimum, we obtained **20 identifiable, 16 weak, and 0 strictly non-identifiable** parameters of 36 at the 95% level, with a 25-fold logarithmic scan window. The agreement with the Fisher analysis is close and, importantly, independent of it. The tightest intervals belong to ε_M (relative width 0.031; [0.596, 0.614] against a true 0.60), ε_{13} (0.053), ε_R (0.060), and λ_P (0.073; [1.576, 1.692] against a true 1.60). The widest belong to η_R (2.336), η_{M13} (2.279), and δ_{P1} (2.144; [4.41, 16.12] against a true 3.50), with W at 0.862 and θ_P at 0.800, the latter with an interval of [0.157, 0.785] against a true value of 0.50.

The parameters with the widest profiles are exactly the ones the Fisher matrix flags as near-degenerate pairs: θ_P with δ_{P1} , and η_R with η_M . Two methods with entirely different assumptions — one a local quadratic expansion at the truth, the other a global re-optimized scan from an estimated optimum — identify the same pairs. That is the strongest evidence available here that the degeneracy is real and not an artifact of either diagnostic.

We must state a limitation squarely: the profile-likelihood analysis completed for **one regime only** (Advanced Adenoma). The remaining three reached the optimization stage and did not finish the profiling within the compute budget, so no confidence intervals were written for them. We therefore do not claim a four-regime profile-likelihood analysis, and the numbers above should be read as a single-regime corroboration of the Fisher result rather than an independent four-regime replication.



7.5 Consensus verdict

The evidence converges. An analytic argument shows that θ_P and δ_{P1} are determined only through their product. The Fisher matrix finds that degeneracy in every regime without being told, and finds the number of additional near-degenerate pairs growing $0 \rightarrow 1 \rightarrow 6 \rightarrow 10$ with WNT drive. Profile likelihood, from a different starting point and with no quadratic approximation, assigns the widest confidence intervals to the same parameters. And the inverse runs of Section 6, using no identifiability machinery at all, recover θ_P in only 5 of 28 attempts and degrade monotonically across the same ordering.

The mechanism is the one anticipated in Section 2. Increasing W drives β -catenin into saturation of the Hill terms that read it; in that limit the APC subsystem becomes insensitive to how much APC functionality remains, and θ_P decouples from the observables. The degeneracy is therefore a conditioning failure created by the operating point, not a flaw in the model structure — which is why the de-saturating $D_W = -0.7$ condition of Section 6 improves matters (recovering θ_P to 0.9% in the Normal regime) without eliminating the problem in the most extreme regime.

8 Bayesian inference and posterior calibration

Everything reported so far is a point estimate accompanied by an *ad hoc* identifiability verdict — a parameter is called recovered if its relative error falls below a threshold. A Bayesian treatment replaces this with a posterior, from which credible intervals and a principled identifiability statement follow directly: a marginal that has collapsed relative to its prior is informed by the data, and one that has reverted to the prior is not [21]. We implemented both halves of this: a Bayesian *inverse* PINN, sampling the parameters with the state networks frozen, and a Bayesian *forward* PINN, sampling the network weights with the parameters known.

We report this section in an unusual amount of detail because our first inverse run failed in a way that is instructive, generic to Bayesian PINNs, and — in our reading of the literature — not adequately flagged. It produced posteriors that were simultaneously extremely confident and badly wrong, and the naive identifiability verdict computed from them declared all 36 parameters identifiable in all four regimes.

8.1 Sampler and priors

Both analyses use Hamiltonian Monte Carlo [42], which is the appropriate sampler here: gradients of the log posterior are available by automatic differentiation at no extra cost, and HMC’s efficiency degrades far more gracefully with dimension than random-walk Metropolis or variational approximations. Our implementation is a static-path HMC with a dense mass matrix rather than NUTS. The mass matrix is initialized from the Laplace approximation at the MAP — the Hessian of the potential, eigenvalue-floored to guarantee positive definiteness — and then re-adapted from the warmup sample covariance. The step size is set by the standard heuristic search followed by dual

averaging to a target acceptance rate of 0.8 [43], and the trajectory length is jittered by ± 3 leapfrog steps per draw to avoid resonances. Non-finite Hamiltonians are hard-rejected as divergences.

Sampling is performed in the unconstrained space in which the parameters are represented: 35 of the 36 are sampled in log space and θ_P in logit space, so positivity and the $(0, 1)$ constraint on θ_P are enforced by construction. The priors are Gaussian in that space, with $\sigma_{\text{prior}} = 1.0$ on the log parameters and $\sigma_{\text{prior}} = 2.0$ on the logit of θ_P , the latter being close to flat on $(0, 1)$. Chains are initialized at the integral-residual MAP estimate of Section 6.6.

We note one limitation up front: each regime is sampled with a single chain, so no $\hat{R}$ diagnostic is available. Mixing is assessed by effective sample size alone, computed by Geyer’s initial-monotone-sequence truncation.

8.2 Likelihood specification and a calibration failure

Because the state networks are frozen, the data term is constant in θ and drops out of the posterior; the only θ -dependent term is the physics residual. The first implementation took the obvious route and treated every residual evaluation as an independent observation:

$$U(\theta) = \frac{1}{2\sigma_{\text{res}}^2} \sum_{i=1}^{N_{\text{res}}} r_i(\theta)^2 + \sum_j \frac{s_j^2}{2\sigma_{\text{prior},j}^2}, \quad (32)$$

with σ_{res} calibrated so that the reduced χ^2 equals one at the MAP, i.e. $\sigma_{\text{res}} = \sqrt{\text{SSR}_{\text{MAP}}/N_{\text{res}}}$.

The trouble is N_{res} . With 10 conditions, 7 states, and 3000 collocation points, $N_{\text{res}} = 209,930$, and σ_{res} calibrates to $\sim 5 \times 10^{-4}$. Nothing in Eq. (32) knows that adjacent collocation points on a smooth trajectory carry almost no independent information. The posterior precision therefore scales with the *density of the collocation grid* rather than with the information content of the experiment: refine the grid and the posterior tightens, without a single new measurement having been made. The potential energy sits around $U \approx 8.7 \times 10^4$, and the resulting posterior is a needle.

The consequences are recorded in Table 11. Effective sample sizes are 3–6 out of 1500 draws, so the chain is not mixing at all; the posterior mean sits a median of 29–51 standard deviations from the truth, with worst cases in the hundreds; and the 95% credible interval covers the true value for at most 1 of 36 parameters in any regime. Individual cases make the failure vivid. In the Severe APC Loss regime the posterior for W is 1.1203 ± 0.0043 against a true value of 2.00 — a z -score of -206 — while θ_P is 0.6134 ± 0.0011 against a true 0.25, a z -score of $+332$.

Regime	ESS (median)	median $ z $	max $ z $	CI ₉₅ covers truth	verdict reported
Normal	3.4	38.4	784	1/36	36/36 IDENT
Early Adenoma	5.6	29.0	466	0/36	36/36 IDENT
Advanced Adenoma	3.1	50.1	293	1/36	36/36 IDENT
Severe APC Loss	5.4	50.9	563	1/36	36/36 IDENT

Table 11: The miscalibrated first run. $z = (\text{posterior mean} - \text{truth})/\text{posterior std}$; ESS is out of 1500 draws. Every diagnostic that was actually computed at the time (posterior width) looked excellent; every diagnostic that was not (mixing, coverage) was catastrophic.

The last column is the important one. The identifiability verdict was computed purely as *shrinkage relative to the prior*,

$$\text{shrink}_j = \frac{\text{std}(s_j | \mathcal{D})}{\sigma_{\text{prior},j}}, \quad \text{IDENT if shrink} < 0.30, \quad \text{WEAK if } 0.30 \leq \text{shrink} < 0.70, \quad \text{NON-IDENT otherwise}, \quad (33)$$

and these posteriors were *tight but biased*, with $\text{shrink}(W)$ in the range 0.0006–0.0038. A width-based criterion reads that as maximal information. The general lesson is that in a Bayesian PINN, posterior width is not evidence of identifiability unless the sampler is mixing and the intervals are calibrated; a physics likelihood summed over a dense collocation grid can manufacture arbitrarily small credible intervals around an arbitrarily wrong answer.

8.3 The corrected likelihood and its effect

The fix is to stop pretending that 2×10^5 correlated residual evaluations constitute 2×10^5 independent observations. We rescale the physics term by a fixed effective sample size:

$$U(\theta) = \frac{n_{\text{eff}}}{N_{\text{res}}} \cdot \frac{1}{2\sigma_{\text{res}}^2} \sum_{i=1}^{N_{\text{res}}} r_i(\theta)^2 + \sum_j \frac{s_j^2}{2\sigma_{\text{prior},j}^2}, \quad n_{\text{eff}} = n_{\text{cond}} \times 7 \times n_{\text{obs}}, \quad (34)$$

where $n_{\text{obs}} = 40$ is the actual per-trajectory observation budget of the sparse-data setting. With 10 conditions this gives $n_{\text{eff}} = 2800$ against $N_{\text{res}} = 279,930$, a downweighting of roughly $100\times$, and — the point of the construction — a posterior width that no longer depends on how finely the collocation grid is drawn. Written this way the physics term is an n_{eff} -weighted *mean* residual rather than a sum, and σ_{res} (still the per-point RMS residual at the MAP) becomes orthogonal to the sample-size choice instead of fighting it.

We additionally gate reporting on two diagnostics that would have caught the original failure: a regime’s verdicts are reportable only if the median ESS is at least 200 *and* the credible intervals cover the truth for at least 33 of 36 parameters. Table 12 gives the corrected run.

Regime	ESS (median)	median $ z $	covers truth	accept	gate
Normal	96	1.3	21/36	0.79	fail
Early Adenoma	1265	1.6	20/36	0.68	fail
Advanced Adenoma	12	5.8	8/36	0.70	fail
Severe APC Loss	15	9.9	9/36	0.67	fail

Table 12: The corrected run (3000 warmup, 3000 draws, 48 leapfrog steps). The posteriors are now approximately honest — median $|z|$ has fallen from 29–51 to 1.3–9.9 — but the gate still fails in all four regimes, so we do not report IDENT/NON-IDENT counts from them.

We emphasize what this table does and does not license. The median $|z|$ has fallen by more than an order of magnitude, from 29–51 to 1.3–9.9, so the posteriors are no longer confidently wrong. But no regime passes the gate, so we do not quote a Bayesian identifiability count as a result. What the table does support is a calibration-based restatement of the θ_P wall that is independent of any threshold: coverage degrades monotonically with WNT drive, $21/36 \rightarrow 20/36 \rightarrow 8/36 \rightarrow 9/36$, and the median $|z|$ climbs monotonically, $1.3 \rightarrow 1.6 \rightarrow 5.8 \rightarrow 9.9$. The posterior becomes progressively less able to bracket the truth exactly as the system moves into the high-WNT regimes, which is what Sections 6 and 7 predict from entirely different evidence.

Scored the same way as every other estimator in this paper — posterior mean within 10% relative error — the corrected posteriors recover **19/20/11/8** parameters across the four regimes. That is at most a few parameters better than the integral-residual point estimate that seeded the chains (17/16/10/7), and it does not clear the ceiling established in Section 6. This is the honest headline for the Bayesian inverse work: it upgrades point estimates to intervals and exposes the calibration question, but it does not by itself recover more of the model.

8.4 Posterior-predictive uncertainty in the forward problem

The forward twin holds the parameters fixed and samples the network weights, so the posterior-predictive spread is uncertainty about the *solution* given sparse noisy observations. This is a harder sampling problem in the trivial sense — the state of the chain is 10,951 network weights rather than 36 parameters — which forces a diagonal mass matrix, but an easier one in the sense that we only care about a low-dimensional functional of the weights.

The setup uses 40 noisy observations at $\sigma = 0.02$, 800 collocation points, 1000 warmup and 1000 draws at 40 leapfrog steps, with a $\mathcal{N}(0, 3^2)$ prior on the weights and three likelihood terms (data, initial condition, physics), each with its scale calibrated to the corresponding residual at an Adam-trained MAP. A useful self-consistency check falls out of that calibration: the data noise scale recovers as $\sigma_{\text{data}} = 1.952 \times 10^{-2}$ against the true injected 0.02.

The first attempt under-covered badly, at mean 95% coverage of 0.73/0.63/0.63/0.56 across the four regimes, because it reported only the *epistemic* band — the spread of the posterior mean function — and compared it against *noisy* observations. The correct comparison adds the observation noise back, $\sigma_{\text{pred}}^2 = \sigma_{\text{epi}}^2 + \sigma^2$, giving the posterior predictive rather than the posterior mean

band. With that correction the two bands separate cleanly and both are calibrated (Table 13): the epistemic band is 0.3–0.6% of state scale and covers the noise-free truth 91–96% of the time, while the predictive band is 6.7–8.0% of state scale and covers the noisy observations 94–95% of the time.

Regime	predictive cov.	rel. predictive width	epistemic cov.	rel. epistemic width
Normal	0.95	0.080	0.96	0.006
Early Adenoma	0.94	0.073	0.91	0.004
Advanced Adenoma	0.94	0.069	0.96	0.003
Severe APC Loss	0.95	0.067	0.96	0.003

Table 13: Bayesian forward PINN. Predictive coverage is against the noisy observations; epistemic coverage is against the noise-free reference. Band widths are relative to state scale. Mean relative RMSE against the reference is 0.001–0.002 in all regimes.

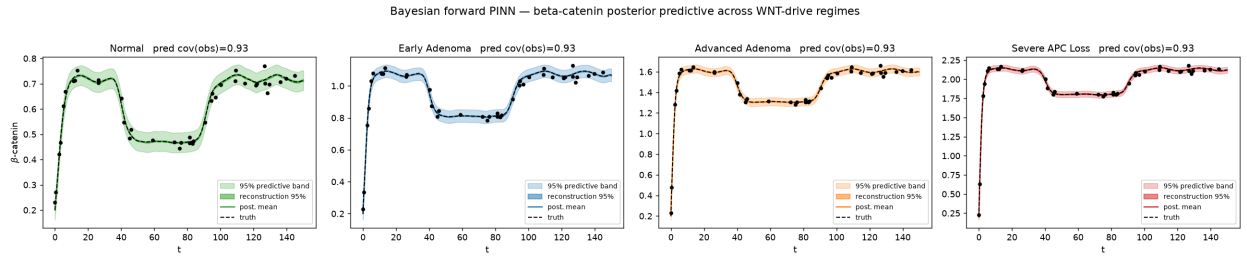


Figure 10: Posterior-predictive bands for β -catenin across the four regimes. Each panel shows the wide 95% posterior-predictive band, the narrow 95% reconstruction (epistemic) band, the posterior mean, the noise-free reference, and the 40 noisy observations. The predictive band is dominated by the observation noise; the epistemic band, which is what the network itself is uncertain about, is roughly an order of magnitude narrower.

Two caveats are worth recording. First, coverage is scored against the same 40 points that entered the data likelihood, so it is an in-sample calibration check, not a held-out one. Second, the ESS of the potential energy itself is 3–4, meaning the weight-space chain is emphatically not mixed; what mixes is the low-dimensional predictive functional, whose ESS is 97–400. The band is therefore trustworthy as a statement about the predicted trajectory and should not be read as a converged exploration of weight space. This is a common situation in Bayesian deep learning and is one reason posterior-predictive quantities are the right thing to report.

9 Discussion

9.1 Principal findings

Read as a whole, the experiments make one argument. For a model of this class, the quantity that limits parameter recovery is the information content of the experiment, and almost nothing else we tried was a substitute for it.

The evidence for that claim is largely negative, which is what makes it credible. Shrinking the network by a factor of five and regularizing it made recovery slightly worse. Alternating two-timescale optimization made it slightly worse. Adding a Stage-3 parameter refinement made no difference. Tripling the number of experimental conditions, when the new conditions perturbed the same input as the old ones, moved recovery from 5 to 7 of 36 while making the estimate of W monotonically worse. Only two interventions produced large gains, and both changed what the estimator was fitting rather than how it searched: replacing the differentiated residual with an integral one (+13 parameters at fixed conditions), and adding perturbation channels that reach the network through new doors (+21 for the classical estimator).

The second of these is the more general lesson, and it is essentially a statement about experimental design. Our first six conditions were all variations of the ATRA protocol, and every one of them entered the WNT/HOX/MYC subnetwork through the single term $\eta_{Rr}/(\kappa_R + r)$. No amount of reshaping a signal that passes through one door will separate the parameters behind it. Four conditions that drive β -catenin and MYC directly were worth more than any number of the former, and the single most targeted of them — a sustained WNT-inhibiting step designed to pull β -catenin out of saturation — recovered θ_P to within 0.9% in the regime where it had previously failed. Designing the perturbation to break a specific degeneracy is a different and more productive activity than collecting more data.

9.2 Methodological implications for physics-informed neural networks

Two findings concern the PINN formulation specifically, and we think both generalize beyond this model.

The autodiff residual is a biased estimator of the parameters. The strong-form residual $\dot{z}_\phi - f(z_\phi; \theta)$ requires the network’s time derivative, and a network can fit a trajectory well while its derivative is noticeably wrong. Any such error is absorbed by θ . Our controlled experiment made the consequence explicit: minimizing the residual to 2.39×10^{-4} — a factor of 1550 below the joint-training value — put W at 37% error, worse than the joint run achieved at a far higher residual. Lower loss, worse parameters. The integral form of Eq. (29) avoids forming the derivative at all and recovers roughly half the gap to the classical estimator. Because published PINN inverse results are almost universally reported with the strong-form residual, this seems worth flagging as a systematic concern rather than a quirk of our system.

The same phenomenon reappeared where we did not expect it, in model selection. Choosing between two restarts by their final physics residual picked the worse-recovering restart in three of four regimes; the better-recovering choices would have yielded 55 rather than 50 parameters. If residual value does not rank two fits of the same model correctly, it should not be used as a selection criterion for inverse problems.

Posterior width is not evidence of identifiability. Our Bayesian inverse PINN initially reported all 36 parameters as identifiable in all four regimes, on the basis of posteriors that had collapsed to a small fraction of their priors. Those posteriors excluded the truth by a median of

29–51 standard deviations, with effective sample sizes of 3 to 6 out of 1500 draws. The cause was treating 2.1×10^5 correlated residual evaluations on a dense collocation grid as independent observations, which makes posterior precision a function of grid density rather than of information. Because the verdict criterion measured only shrinkage relative to the prior, a tight-and-wrong posterior read as maximal information.

We suspect this failure is common and under-reported, for a structural reason: a Bayesian PINN’s physics likelihood has no natural sample size, the number of collocation points is chosen for numerical accuracy rather than statistical reasons, and the diagnostic that would catch the problem (interval coverage against known truth) is unavailable in the applied setting where the truth is unknown. Our mitigation — rescaling the physics term by a fixed effective sample size tied to the real observation budget — restores grid-independence and brought the median discrepancy down from 29–51 to 1.3–9.9 standard deviations. It did not, however, produce chains that pass our own mixing and coverage gate, so we report no Bayesian identifiability counts. We would rather report a gate that fails than a number that looks clean.

9.3 Biological interpretation

Within the limits of an idealized model fitted to synthetic data, two results have biological content.

The forward solutions predict a **progressive loss of ATRA responsiveness** along the adenoma–carcinoma sequence, and they predict it unevenly across species. HOXA5 responds strongly to treatment in all four regimes, but HOXA13 barely falls at all in Advanced Adenoma and Severe APC Loss, and APC becomes markedly less responsive because the elevated degradation $\delta_P(\theta_P)$ caps the level it can reach regardless of how much HOXA5 is induced. The differentiation signal still arrives; the machinery that would translate it into suppression of the proliferative program no longer can. This emerges from the network structure rather than being imposed, and it is consistent with the clinical observation that retinoid monotherapy has been far more successful in acute promyelocytic leukemia [8] than in solid tumors of the colon.

The sensitivity analysis places **APC functionality among the most influential parameters**, with each state otherwise governed by its own pathway driver. Locally it dominates outright, with an elasticity of -3.61 on stemness against 1.04 for the runner-up; measured as a variance share over a range it sits in a leading cluster with the $\text{HOXA13} \rightarrow \beta\text{-catenin}$ and $\text{RA} \rightarrow \text{HOXA5}$ arms rather than standing alone. Either way its prominence is consistent with APC loss being the initiating event of the sequence [1, 3]. The uncomfortable corollary is the one this paper spends most of its length on: the single most influential parameter in the model is also among the least recoverable, recovering in 5 of 28 attempts, precisely because the regimes in which it matters most are the regimes in which β -catenin saturation hides it.

9.4 Limitations

We are working with synthetic data generated from the model itself, with Gaussian noise and all seven states observed. Real measurements would be partial, sparser, differently distributed, and subject to model misspecification of a kind entirely absent here. Every recovery count in this paper should therefore be read as an upper bound.

The parameter values are one internally consistent realization rather than a set fitted to experimental data, so the model is a testbed for methodology and not a calibrated predictive instrument. Its predictions about ATRA responsiveness are qualitative.

On the computational side, several limitations are worth naming. All runs use a single seed, so we report no seed-to-seed variability and no error bars on any count. The ablation that would separate the integral residual from its three accompanying changes did not complete, so that gain is attributed to the bundle. The profile-likelihood analysis completed for one regime of four. The Bayesian chains are single-chain, so no $\hat{R}$ diagnostic is available, and none passes our own reporting gate. No symbolic structural identifiability analysis was performed, so apart from the (θ_P, δ_{P1}) lumping every identifiability claim here is a practical one. And the 27-state sensitivity analysis mean-imputed roughly 8% of failed solves, with a Morris nonlinearity flag that is saturated at that sample size and therefore uninformative.

9.5 Outlook

The obvious next step follows from the finding that information is the binding constraint: rather than adding conditions and checking what improves, use the Fisher matrix or the profile likelihood to *design* the condition that maximally reduces uncertainty along a specific degenerate direction. Our $D_W = -0.7$ de-saturating step was chosen by exactly this reasoning, informally; formalizing it as optimal experimental design is a natural continuation.

A second direction concerns the Bayesian analysis. The correct effective sample size for a physics likelihood is not obvious — ours is a defensible choice, not a derived one — and we suspect a principled treatment, perhaps via the correlation length of the residual field, would be of general use to the B-PINN literature.

Finally, this paper deliberately holds the model structure fixed and asks only about its parameters. The complementary question is what happens when part of the structure is unknown and is replaced by a learned term, as in neural-mechanistic hybrids and universal differential equations [44–46]. That extension raises its own identifiability questions — in particular, whether a learned term absorbs parameters of the equation that hosts it — and is the subject of separate work.

10 Conclusions

This paper makes two claims, one about a model and one about its approximation.

First, we developed a new ODE model of the WNT–RA–HOX axis in colorectal cancer. The system couples seven states — β -catenin, APC, HOXA5, HOXA13, MYC, retinoic acid, and CYP26A1 — through a single elementary principle, that each species accumulates at the rate at which it is produced minus the rate at which it is removed (Eq. (11)). Reducing a 27-state reaction network to these seven, and nondimensionalizing, yields 36 dimensionless parameters and a forcing that superimposes circadian dietary retinoid intake on a smooth ATRA treatment pulse. Four regimes obtained by raising WNT drive W and lowering APC functionality θ_P reproduce the adenoma–carcinoma sequence, and a derived stemness index provides a single scalar readout. The model behaves as a model of this system should: a three-method global sensitivity analysis places APC functionality among the most influential parameters — dominant locally, and in a leading cluster of three when measured as a variance share — gives each state an interpretable own-pathway driver, and shows the reduced system to be nearly additive, in contrast to the strongly interacting network it descends from. Forward solutions predict a progressive and species-specific loss of ATRA responsiveness along the sequence, an emergent consequence of the network structure rather than an imposed assumption.

Second, we established that this system can be approximated by neural networks in both directions. In the forward direction the approximation is accurate and cheap in data: a Fourier-feature network reproduces the stiff Radau reference trajectories to 1–3% relative error across all four regimes, and does so from 40 sparse random observations by leaning on the physics residual, at roughly $2.4\times$ the error of a dense-label fit that uses $2.5\times$ more data. The multi-timescale retinoid forcing makes the Fourier embedding mandatory rather than cosmetic: without it the network recovers under half the treatment-pulse amplitude and none of the circadian ripple. In the inverse direction the approximation also succeeds, but only up to a ceiling that we are able to decompose. An estimator defect — the bias in the automatic-differentiation residual — costs 13 parameters and is removed by an integral residual that never forms the derivative. An information deficit costs 21 more and is addressed only by perturbations that reach the network through new channels; further experiments of the kind already performed are close to worthless.

The residue is a genuine limit of the problem rather than of the method, and four independent analyses agree on where it lies. An analytic argument, the Fisher information spectrum, profile likelihood, and Bayesian posterior coverage all show that as WNT drive increases β -catenin saturates, θ_P decouples from the observables, and the number of near-degenerate parameter pairs grows $0 \rightarrow 1 \rightarrow 6 \rightarrow 10$. Restricting to the eight best-constrained parameters gives a well-posed problem with condition number 10^2 – 10^3 instead of 10^7 – 10^{19} , which locates precisely where the recoverable information lives. The uncomfortable corollary is that the most influential parameter in the model is among the least recoverable, for the same reason in both cases.

Two cautions accompany these results, and we report them with the same weight as the successes. The default proportional sampling box of the sensitivity analysis pushed one parameter past a sign change in the model, producing a variance share above unity and a plausible-looking but incorrect ranking. And a Bayesian PINN whose physics likelihood summed over a dense col-

location grid produced posteriors that were extremely confident and badly wrong, certifying all 36 parameters as identifiable in all four regimes; posterior width is not evidence of identifiability unless the chain is mixing and the intervals are calibrated. We believe the second failure mode is generic to Bayesian PINNs and inadequately flagged in the literature.

Taken together, the model of Section 2 and the approximation results that follow it establish that a mechanistic description of WNT–RA–HOX signaling in colorectal cancer is tractable to physics-informed learning in both the forward and the inverse direction, and they identify the binding constraint on what such learning can deliver. That constraint is the information content of the experiment, not the expressiveness of the network — which makes experimental design, rather than architecture search, the productive direction for work of this kind.

Data and code availability

All code, run logs, trained checkpoints, and numeric result tables are in the project repository. The forward accuracy figures of Table 5 are regenerated from saved checkpoints against a fresh Radau reference and written to `forward_error_table.{txt,json}` in each run directory. The sensitivity tables are written as CSV alongside the figures, in both the original and the θ_P -clipped sampling boxes described in Section 4. Fisher matrices, eigenspectra, correlation matrices and per-parameter verdicts are saved per regime, as are the profile-likelihood confidence intervals and all posterior summaries. Every training run records the random seed, which is fixed at 42 throughout.

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Computations were performed in PyTorch [47] and SciPy [28], with sensitivity analysis by SALib [33].

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